

FFF

Final

Segment - 1

field Effect Transistor (FET)

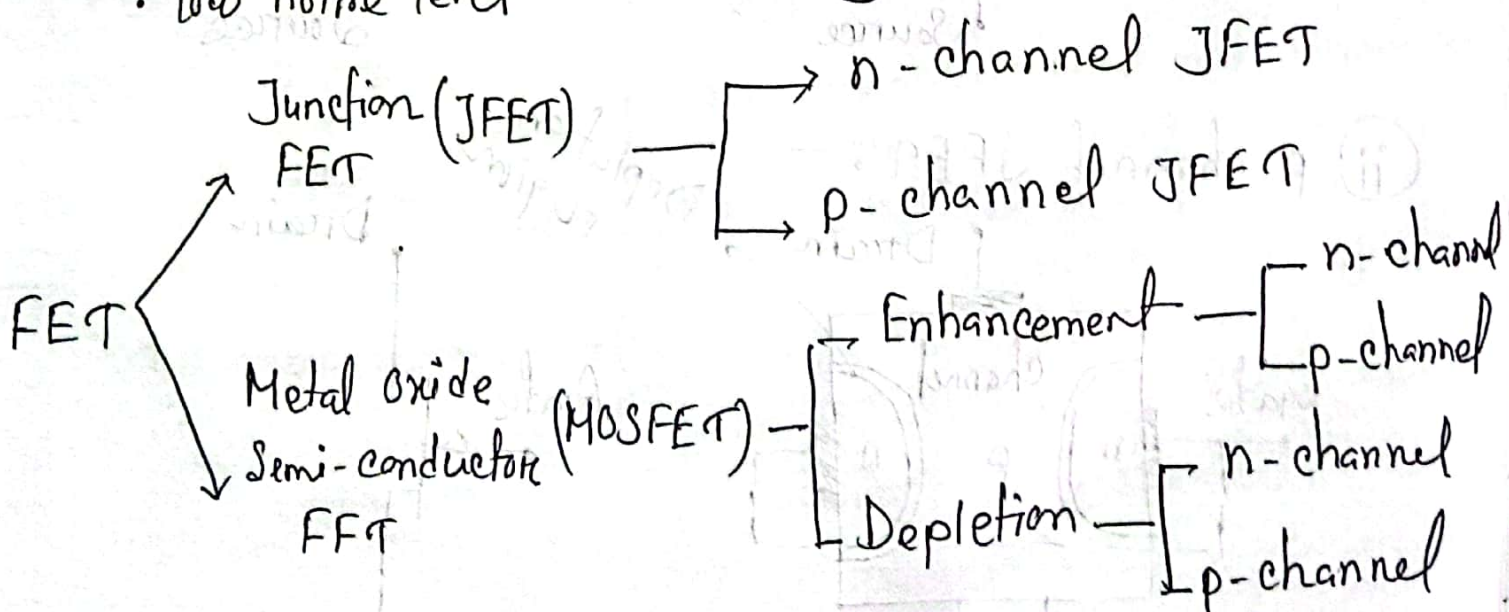
⊛ The field effect transistor is a three terminal unipolar semiconductor device which commonly used for weak signal amplification.

The ordinary/bipolar transistor has two main disadvantages.

- It has a low input impedance.
- It has considerable noise level.

but in FET

- High input impedance
- low noise level the ordinary transistor.

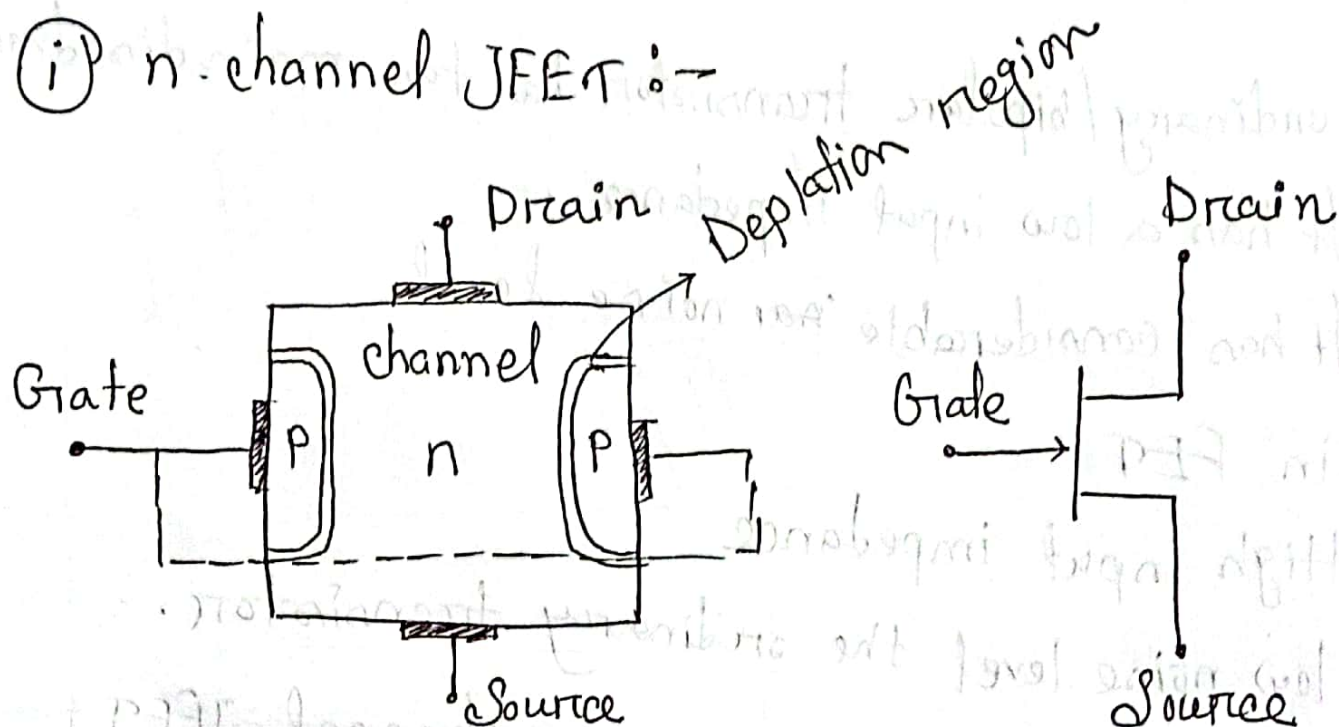


* JFET :

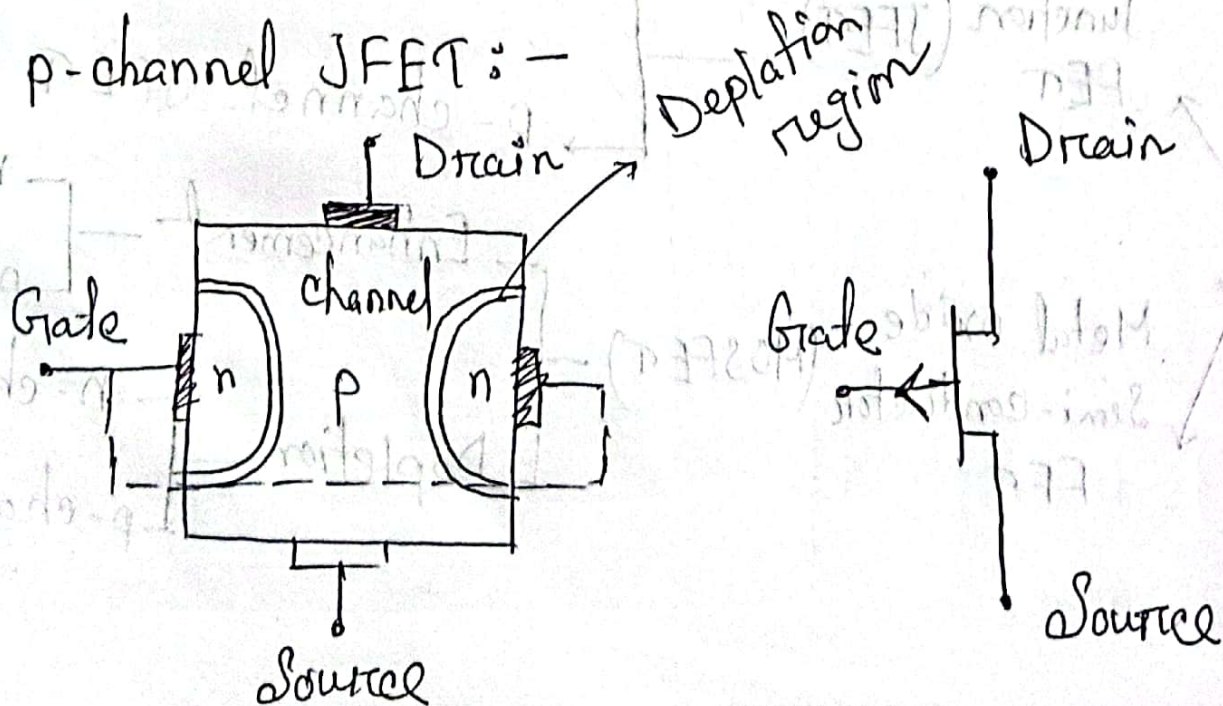
Junction FET is a three terminal semi-conductor device in which current conducted by one type of carrier i.e electron or hole.

Type :-

(i) n-channel JFET :-



(ii) p-channel JFET :-



⇒ Source :- The terminal through ~~which~~ ^{where} the majority carriers enter in the channel.

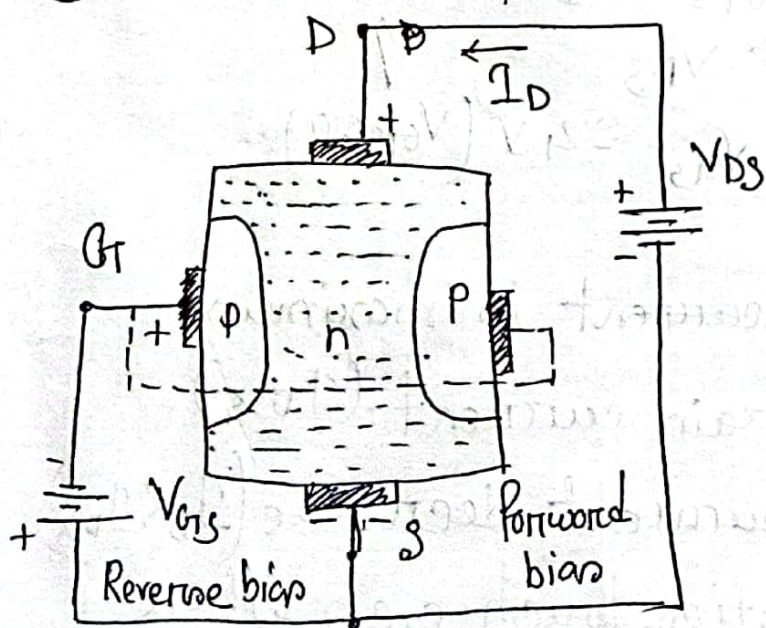
⇒ Drain :- The terminal through which the majority carriers leave from the channel.

⇒ Gate :- Two internally connected heavily doped impurity regions to create two PN junction.

⇒ channel :- The region between source & drain, sandwiched between two gate.

Theory of operation :-

(i) n-channel working operation :-



⇒ Source and drain connected with forward bias -
Note :- when $V_{GS} = 0$
So depletion region is normal.

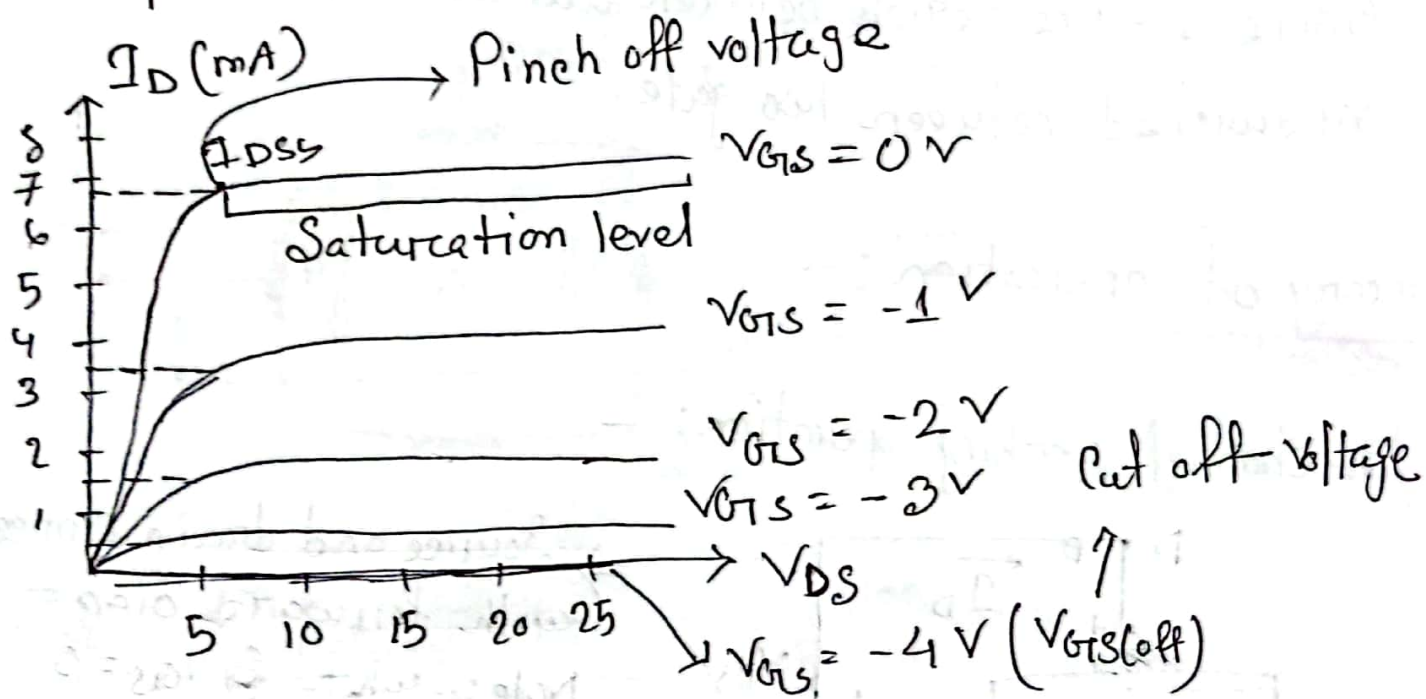
the current flow. ~~flow~~
and this flow is given
the maximum current
($V_{GS} = 0V$) → $I_{D \text{ maximum}}$

~~I_{DSS}~~ I_{DSS}

$\Rightarrow$ When V_{GS} increase, depletion layer are also increase and channel will be decrease.
 and I_D current also decreasing.

~~when at the V_{GS} voltage when~~
 when depletion layer both touch each other
 the current I_D become 0.

Output characteristic curve

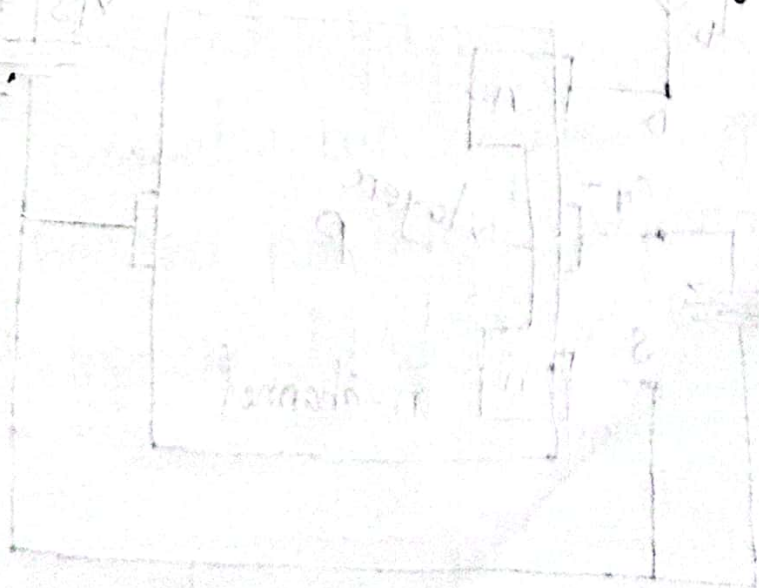


- when $V_{GS} = 0$, the drain current is maximum it called shorted gate drain current (I_{DSS})
- If V_{GS} increase drain current decrease ($I_D < I_{DSS}$)
- The V_{GS} voltage where depletion touch each other called is $V_{GS(off)}$.

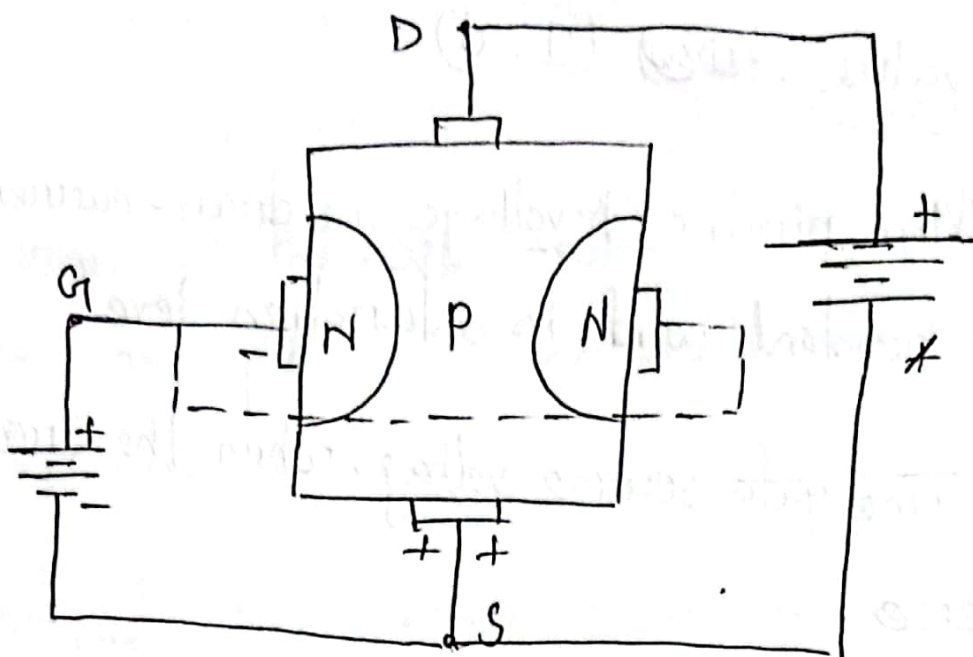
- Pinch off voltage (V_p) $\rightarrow$ It is the minimum drain source voltage at which the drain current essentially become constant saturation value. ($I_D = I_{DSS}$)
- Saturation level $\rightarrow$ After pinch off voltage the drain current become constant, this constant point is saturation level
- Cut off voltage $\rightarrow$ The gate-source voltage, when the drain current become zero.

Feature of JFET :-

- ① JFET is a voltage controlled device i.e input voltage (V_{GS}) control output current (I_D)
- ② ~~It~~ operated with gate-source p-n junction in reverse bias, because of this it has high input impedance.

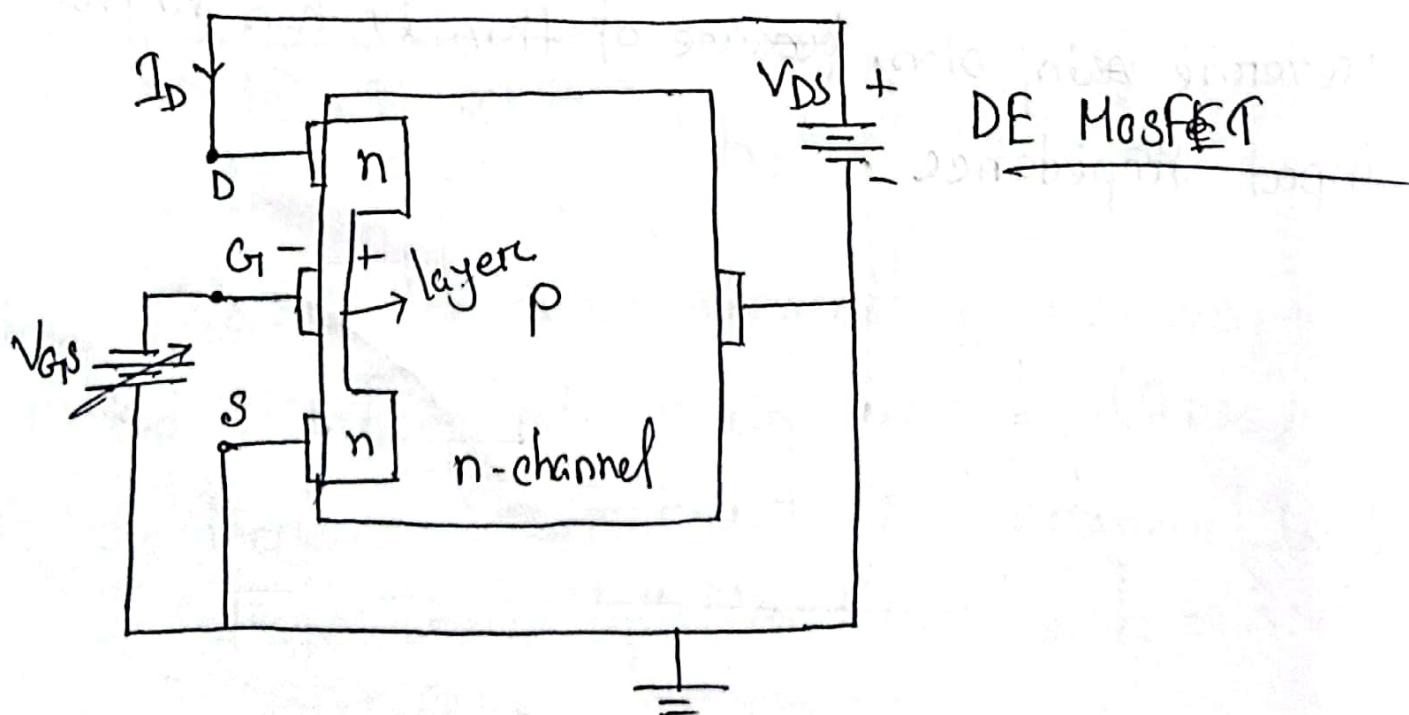


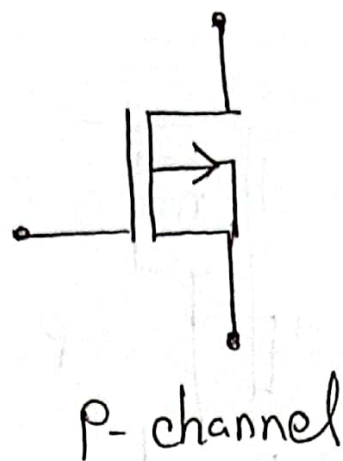
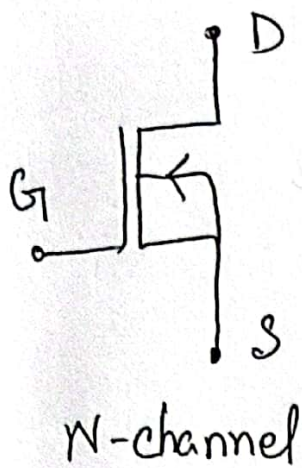
② p-channel circuit :-



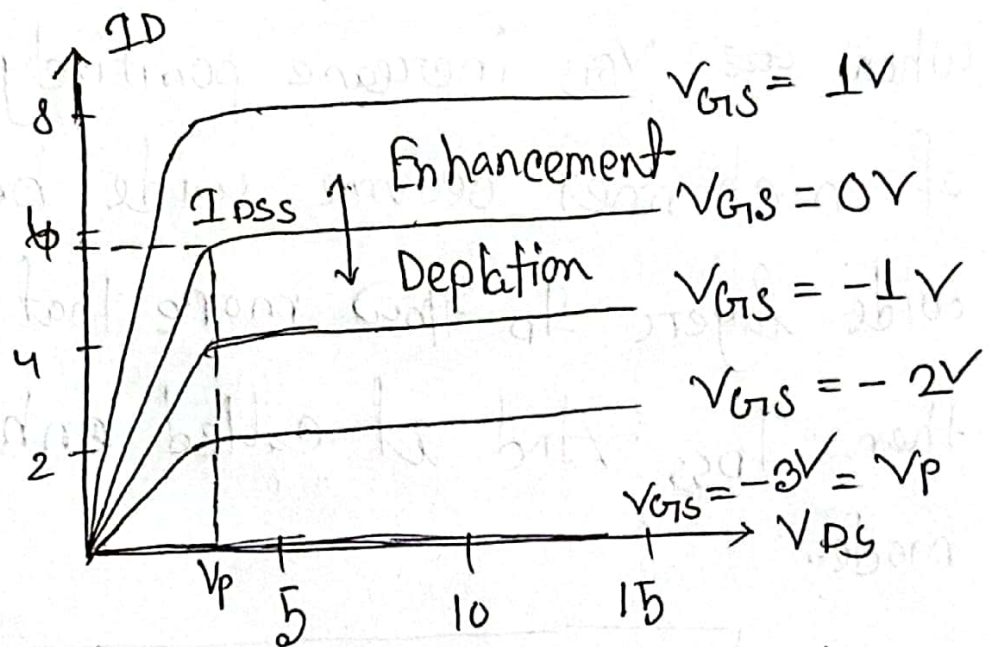
③ MOSFET :-

The MOSFET transistor is a semiconductor device which is widely used for switching and amplifying electronic signals in the electronic devices.

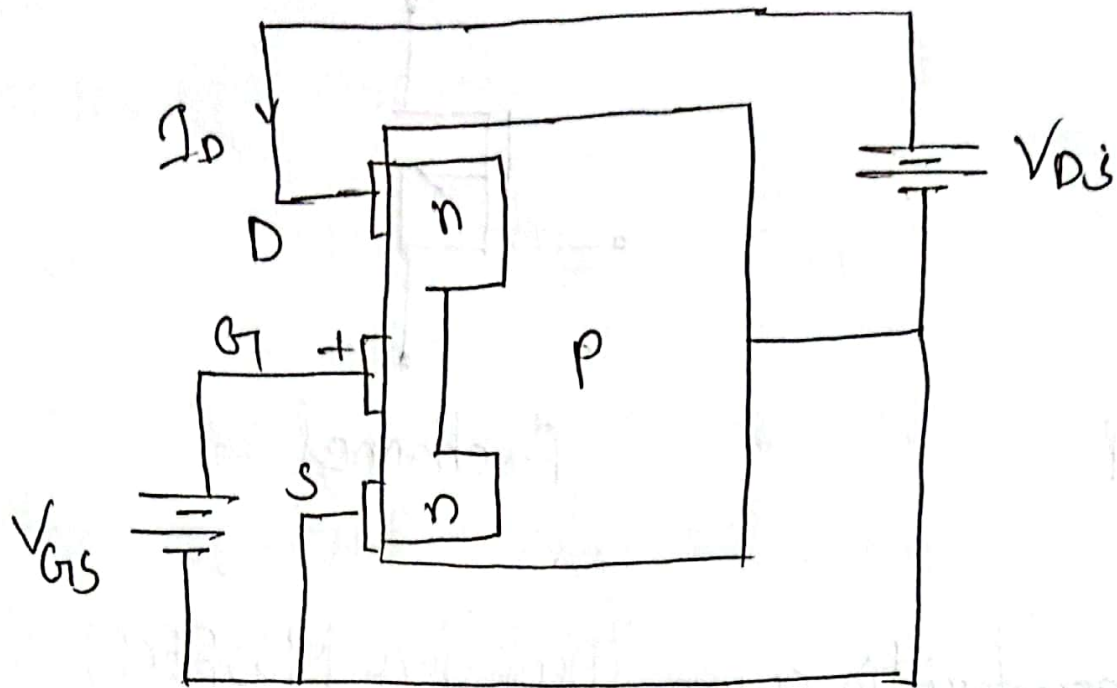




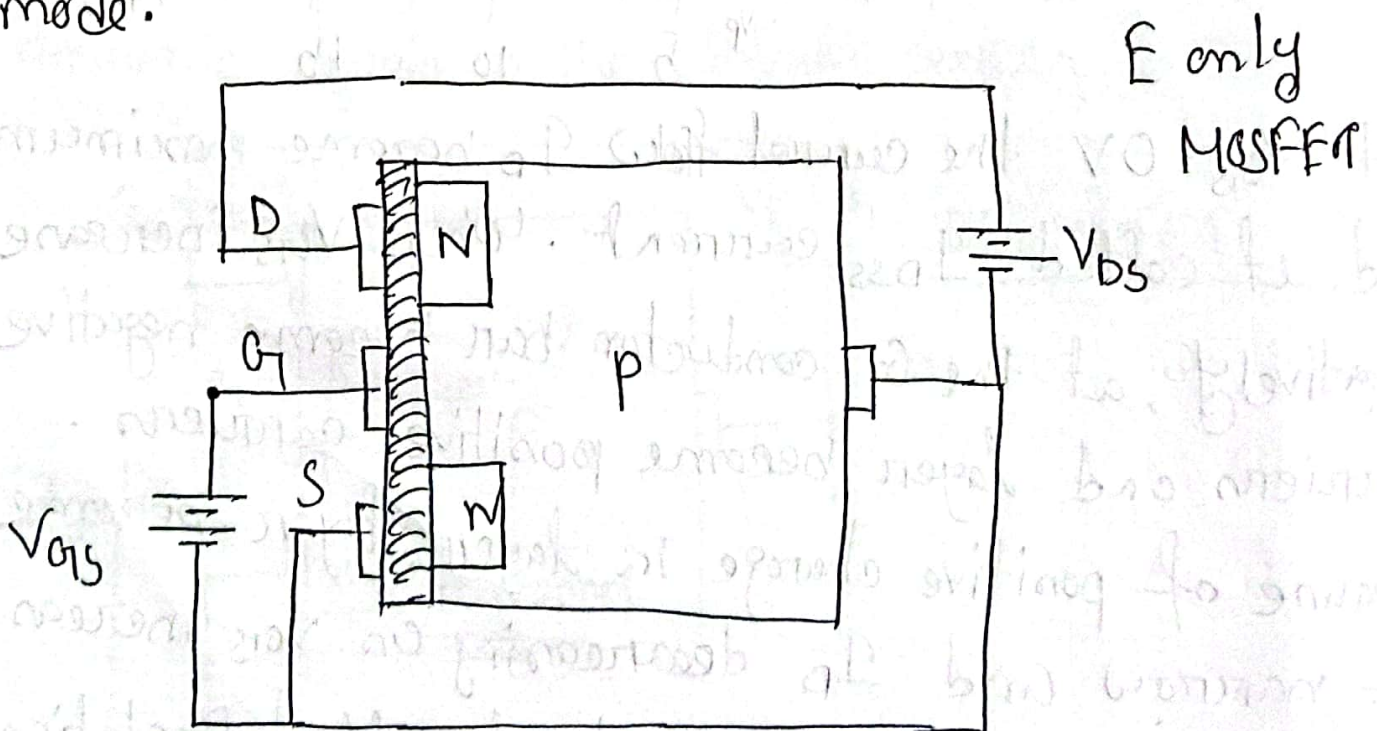
Output characteristic curve (Depletion MOSFET)

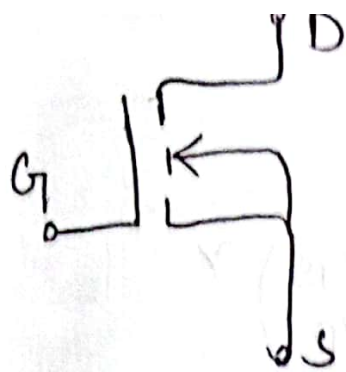


At $V_{GS} = 0V$ the current I_D become maximum and it called I_{DSS} current. When V_{GS} increase negatively, at the G1 conductor bar become negative carriers and layer become positive carriers. because of positive charge in layer, layer become narrower and I_D decreasing as V_{GS} increase V_{GS} become V_P , $I_D = 0A$ and it called Depletion mode

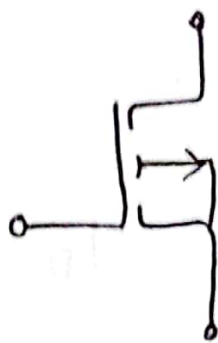


When ~~use~~ V_{GS} increase positively, the layer of n-channel become wide, because of wide layer I_D flow more that it is more than I_{DSS} . And it called enhancement mode.



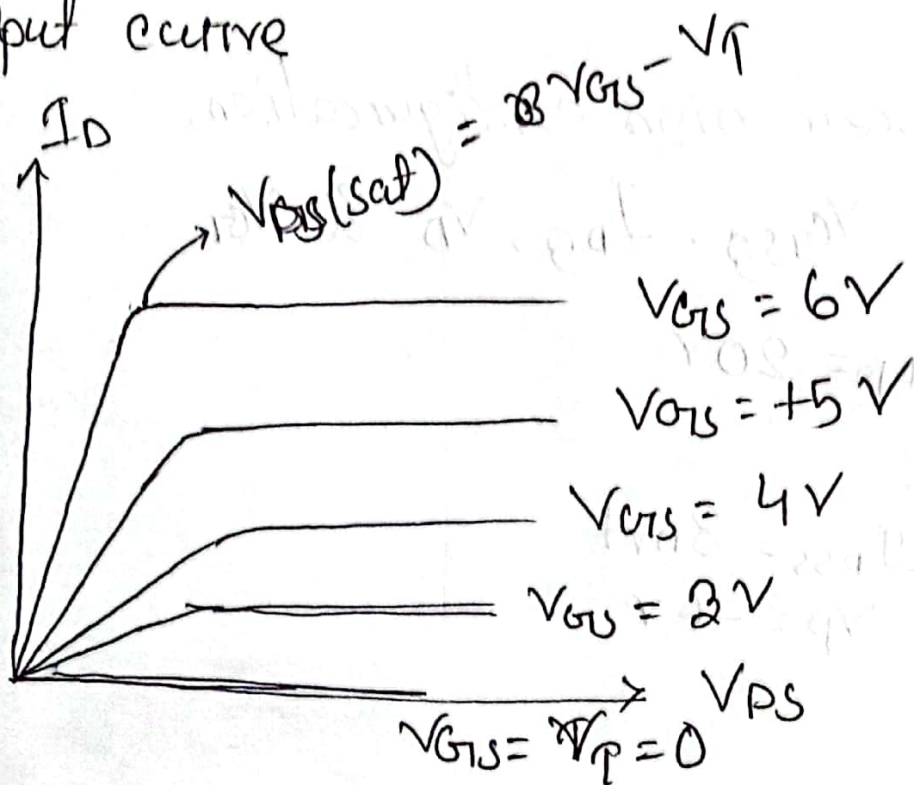


N-type



P-type

Output curve



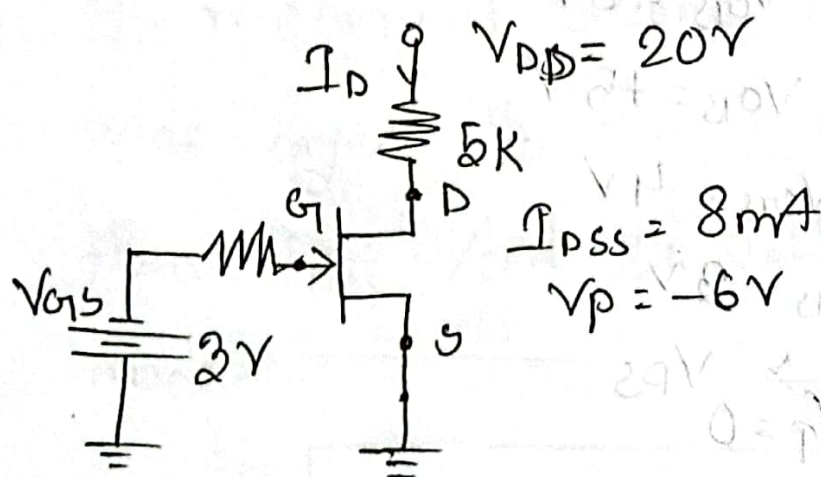
* Math :- JFET

Shockley Equation :- $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$

Note :- $I_D = I_{DSS}$ [when $V_{GS} = 0$]

$I_D = 0$ [when, $V_{GS} = V_P$]

Ex :- For the fixed bias configuration, determine : V_{GSQ} , I_{DQ} , V_D & V_{GT}



Sol :- Here gate current is zero

$\therefore V_{GSQ} = -2V$ Ans

$\therefore I_{DQ} = I_{DSS} \left(1 - \frac{V_{GSQ}}{V_P}\right)^2$

$= 8 \left(1 - \frac{-2}{-6}\right)^2$

$= 2mA$ Ans

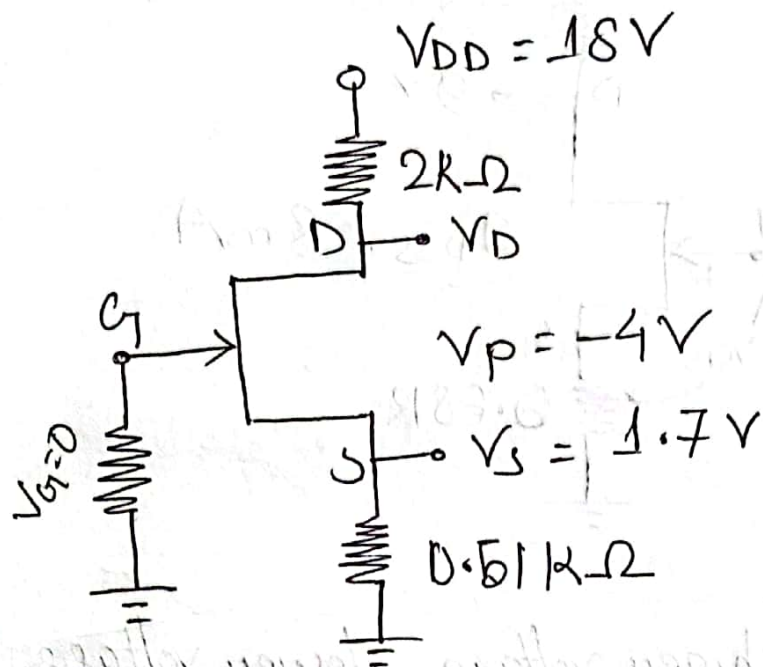
$V_D / V_{DS} = V_{DD} - (5 \times I_D)$

$= 20 - (5 \times 2)$

$= 10V$ Ans

$V_{GT} = V_{GS} = -2V$ Ans

Ex:- For the self bias configuration Determine I_{DQ} , V_{GSQ} , I_{DSS} , and V_{DS}



$$I_D = I_S$$

Sol :- $I_{DQ} = I_S = \frac{V_S}{R} = \frac{1.7}{0.51} = 3.33 \text{ mA}$ Ans

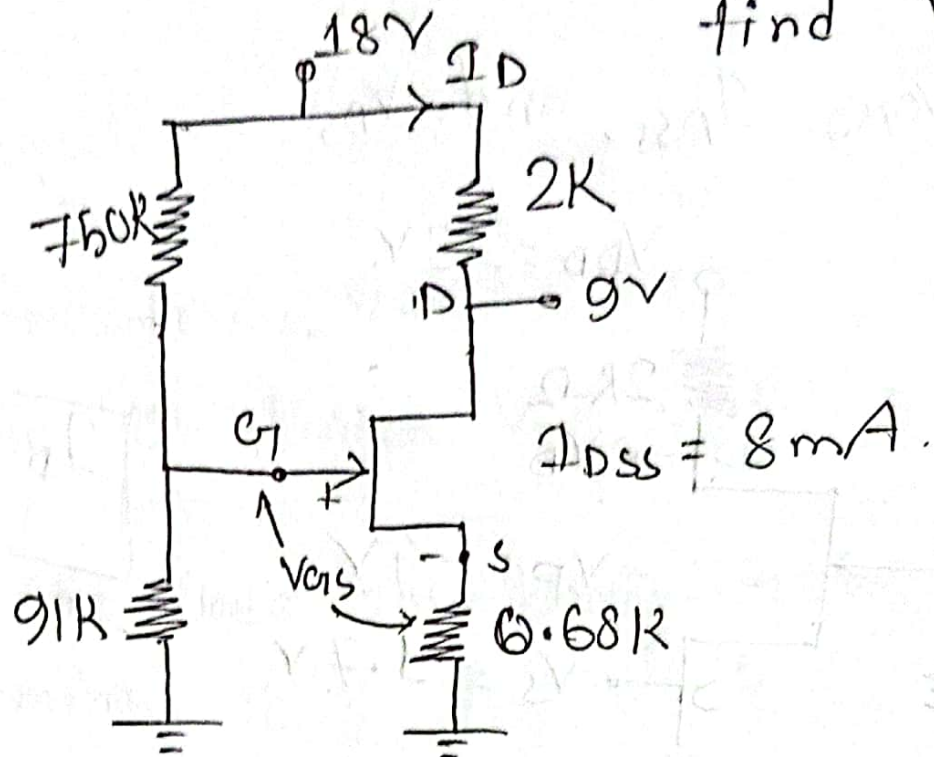
$V_{GS} = V_G - V_S = 0 - 1.7 \text{ V} = -1.7 \text{ V}$ Ans

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$\Rightarrow I_{DSS} = \frac{I_D}{\left(1 - \frac{V_{GS}}{V_P} \right)^2} = 10.08 \text{ mA}$ Ans

$V_{DS} = V_{DD} - (I_D \times R) - V_S = 18 - (2 \times 3.33) - 1.7$
 $= 9.63 \text{ V}$ Ans

Ex :- The voltage divider configuration find V_{GS} and V_P



Sol :-

$$I_D = \frac{V_{DD} - V_P}{R} \quad \text{higher voltage - lower voltage}$$

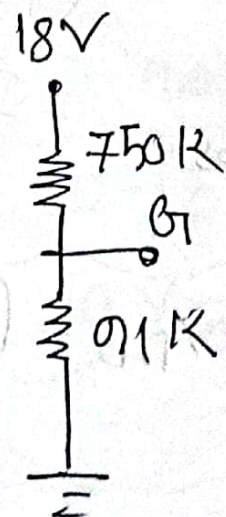
$$= \frac{18 - 9}{2} = 4.5 \text{ mA} \quad \underline{\underline{Ans}}$$

$$\therefore I_D = I_S$$

$$\therefore V_S = I_S R_S = 4.5 \times 0.68 = 3.06 \text{ V} \quad \underline{\underline{Ans}}$$

$$V_G = \frac{18}{750 + 91} \times 91 = 1.948 \text{ V} \quad \underline{\underline{Ans}}$$

$$V_{GS} = V_G - V_S = 1.948 - 3.06 = -1.112 \text{ V} \quad \underline{\underline{Ans}}$$



$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$\Rightarrow 1 - \frac{V_{GS}}{V_P} = \sqrt{\frac{I_D}{I_{DSS}}}$$

$$\Rightarrow V_P = \frac{V_{GS}}{\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)} = \frac{-1.112}{1 - \sqrt{\frac{4.5}{8}}} = \underline{\underline{-4.448V}}$$

~~*Q1 Path of MOSFET~~

MOSFET

$$(1) \quad k = \frac{I_D}{(V_{GS} - V_T)^2} \Rightarrow I_D = k(V_{GS} - V_T)^2$$

Segment - 2

Switching circuit and Multivibrators

- ⇒ Introduction of switching circuit
- ⇒ Type of switching circuit
- ⇒ Switching transistors -
- ⇒ Switching action of a transistor
- ⇒ Introduction of Multivibrator & its type
- ⇒ B Transistor astable and monostable Multivibrator
- ⇒ Schmitt trigger.

□ Switching circuit :-

A circuit which can turn ON or OFF current in an electrical circuit is known as switching circuit.

There are three types of switching circuit

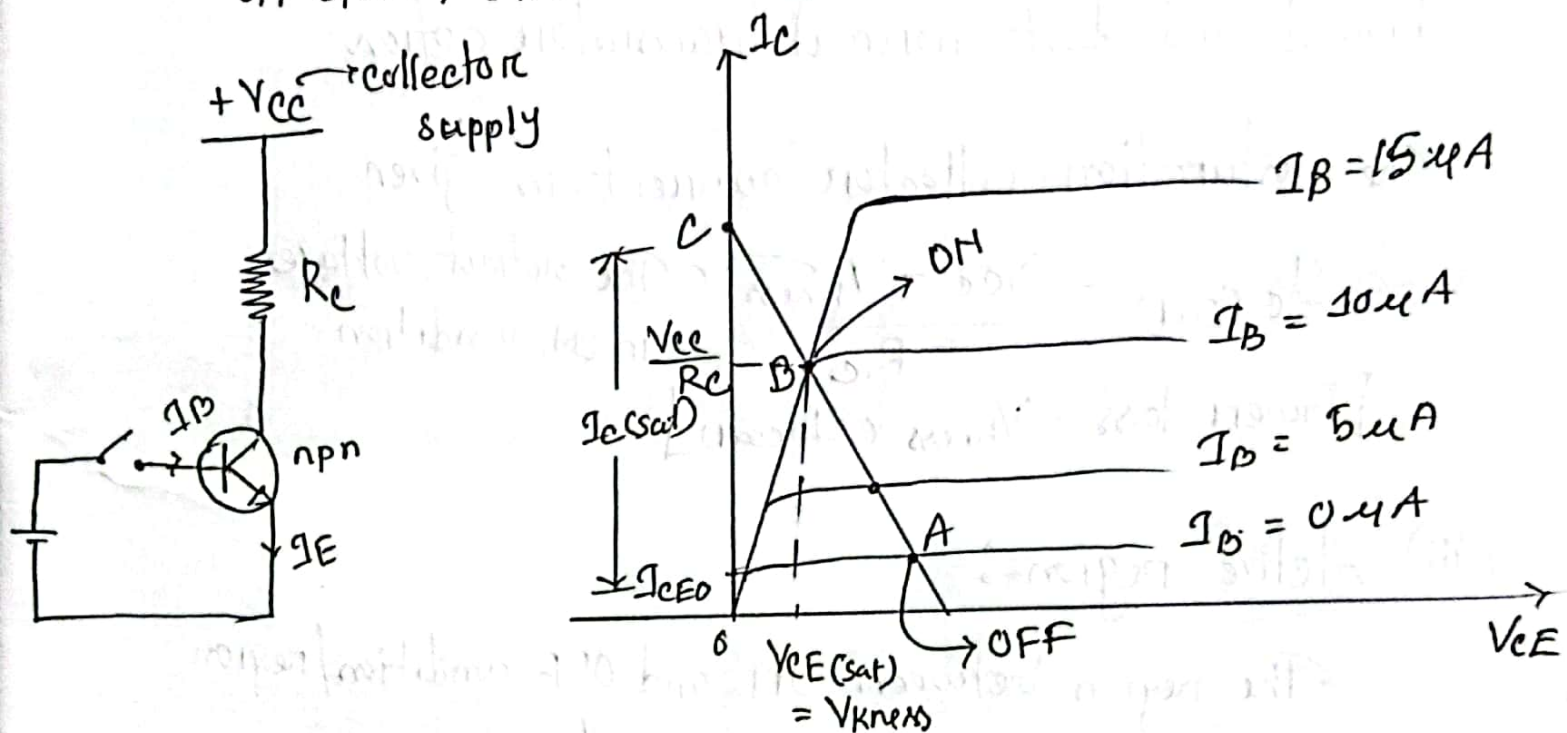
- ① Mechanical switch
- ② Electro-mechanical switch or Relay
- ③ Electronic switch

* Switching Transistors :-

A transistor which is used as a switch is known as switching transistor.

Note: Maximum current = Saturation collector current
Minimum current = Collector leakage current

ON state $\rightarrow$ collector saturation current flow through load.
OFF state $\rightarrow$ Collector leakage current flow through load.



i) OFF region :- (Cut off)

When the input base voltage is zero or negative, transistor said to be in the OFF condition. In this condition, $I_B = 0$ and only small collector leakage current (I_{CEO}) exists. Base-emitter junction no longer remain forward and it lost its action and $V_{CE(cutoff)} = V_{CC}$ [Power loss $\approx V_{CC} \times I_{CEO}$]

(ii) ON region :- (Saturation)

When input voltage is made so much positive that saturation collector flows, transistor said to be ON condition. Base current is maximum, so ~~it~~ is the collector current. At this, Collector-Base junction is no longer remain reverse biased and lost normal transistor action.

The saturation collector current is given

$$I_c(\text{sat}) = \frac{V_{CC} - V_{K_{\text{ness}}}}{R_c}$$

The output voltage in ON condition

$$[\text{Power loss} = V_{K_{\text{ness}}} \times I_{c(\text{sat})}]$$

(iii) Active region :-

The region between ON and OFF condition/region is known as active region. In this region collector-base junction remain reverse and base-emitter junction remain forward biased. So transistor will function normally in this region.

Multivibrator :-

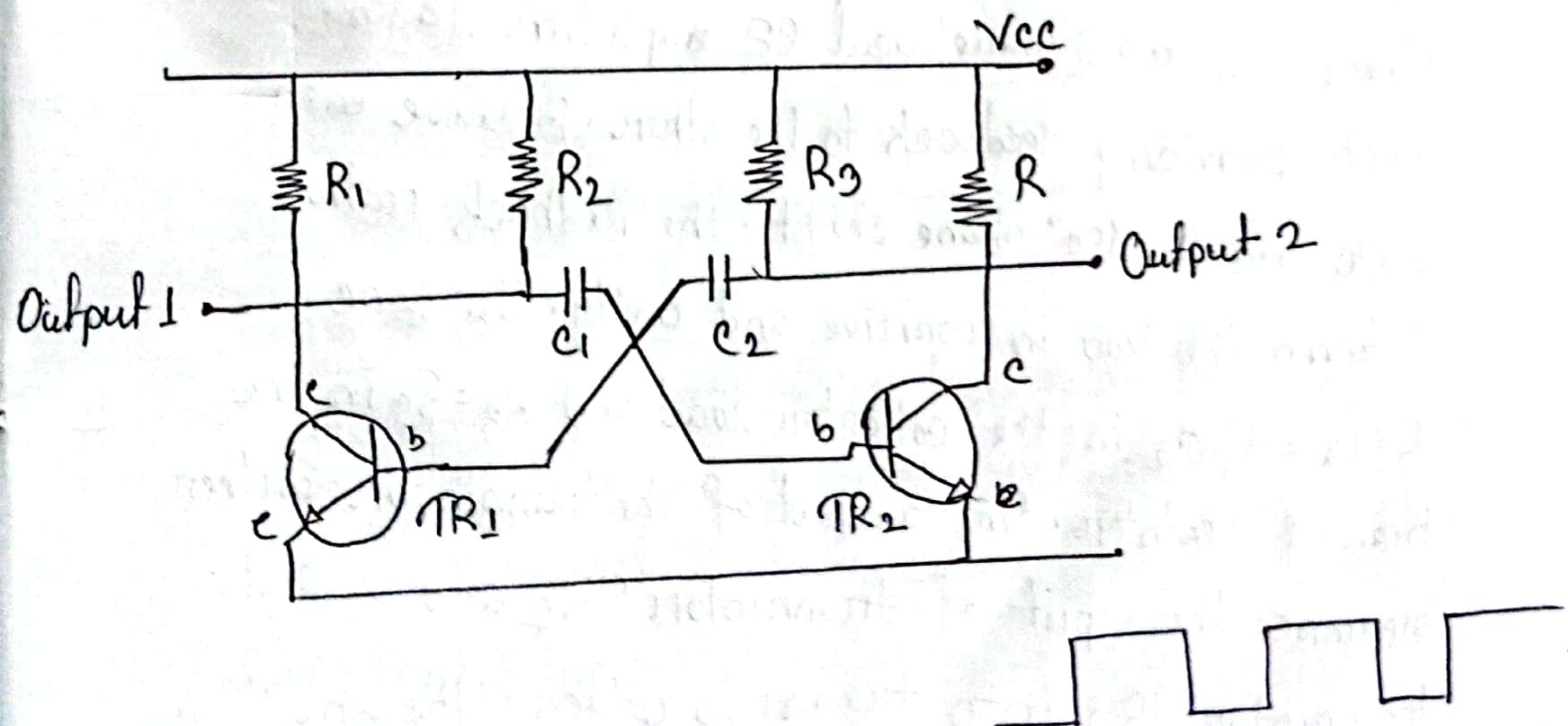
An electrical electronic circuit that generates square wave is known as a multivibrator.

It has also three type —

- ① Astable or free running multivibrator
- ② Monostable or one-shot multivibrator
- ③ Bi-stable or flip-flop multivibrator.

* Transistor Astable Multivibrator :-

A multivibrator which generates square waves of its own is known as an astable multivibrator.



The basic transistor circuit for an Astable Multivibrator produces a square wave output from a pair of grounded emitter cross-coupled transistors.

Circuit detail

Here using two identical transistors TR_1 and TR_2 . Collector loads of the two stages are equal i.e. $R_1 = R_4$ and biasing resistances are also equal i.e. $R_2 = R_3$. The output TR_1 is coupled to the input TR_2 through C_1 and output TR_2 is coupled to the input TR_1 through C_2 . The wave output can be taken from TR_1 or TR_2 .

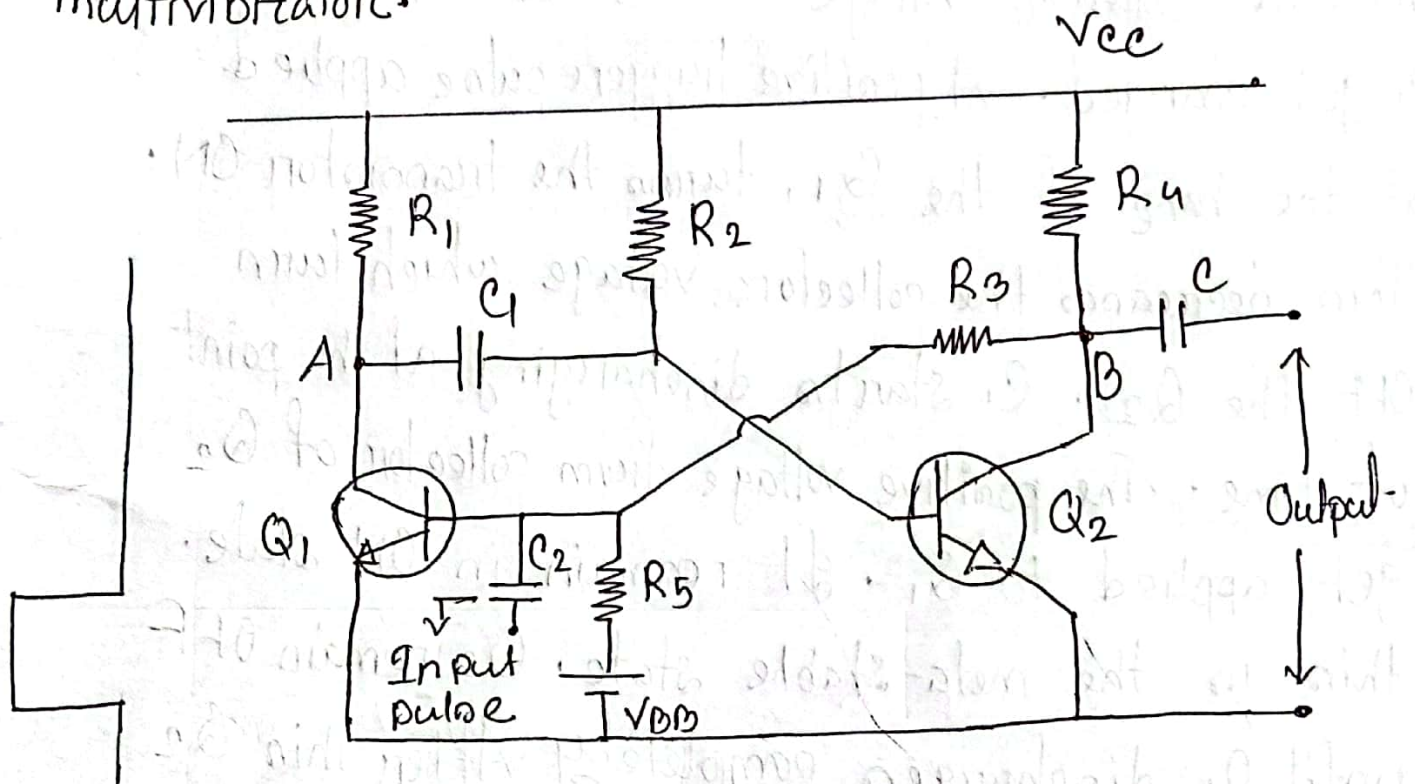
Operation

There are two symmetrical CE amplifier stages, each providing feedback to the other. Because each stage has a 180° phase shift, the feedback ratio between the two is positive and unity. In case $R_2 C_1 = R_3 C_2$ is the collector load and $R_2 = R_3$ is the biasing resistor. The output of the transistor further increases the input of transistor TR_2 and output transistor TR_2 into transistor TR_1 . TR_1 and TR_2

Supply the square wave output by driving the transistors to saturation or cut-off.

* Transistor Monostable Multivibrator:-

A multivibrator which one transistor always conducting and other is non-conducting is called monostable multivibrator.



Circuit detail:

It consists of two similar transistors Q_1 , Q_2 with equal collector load i.e. $R_1 = R_4$. The value of V_{BB} and R_5 are such as to reverse bias Q_1 and keep it at cut off. The collector supply V_{CC} and R_2 forward bias Q_2 and keep it at saturation. The input pulse is given

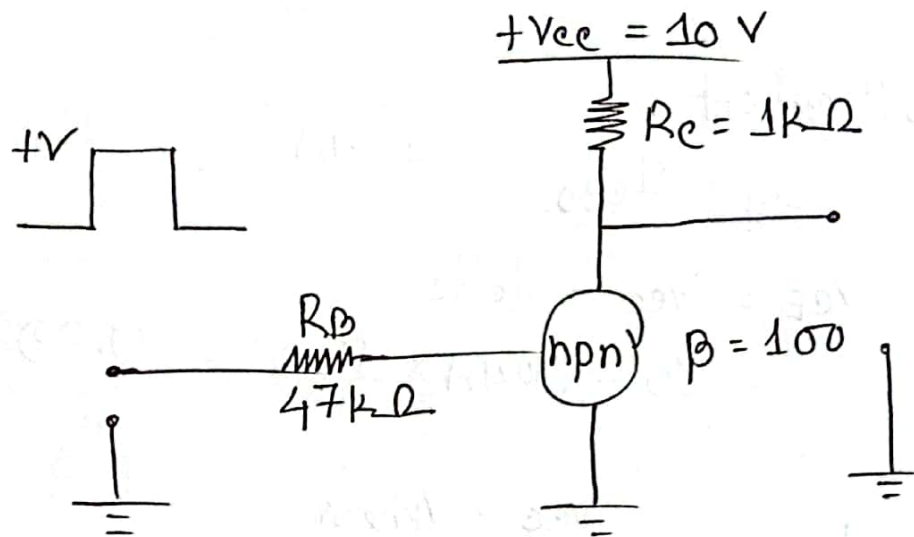
through C_2 to obtain the square wave. Again output can be taken from Q_1 or Q_2 .

Operation

When circuit switched ON, Q_1 will be in OFF state and Q_2 will be in ON state [Stable state]. As Q_1 is OFF, the collector voltage will be V_{cc} at point A and C_1 get charged. A positive trigger pulse applied at the base of the Q_1 , turns the transistor ON. This decreases the collector voltage which turns OFF the Q_2 . C_1 starts discharging at the point of time. The positive voltage from collector of Q_2 get applied to Q_1 . It remains in ON state. This is the meta-stable state. Q_2 remains OFF until C_1 discharges completely. After this Q_2 turns ON with voltage applied through the capacitor discharge. This turns Q_1 OFF which is previous stable state.

Math

- ① Determine the minimum high input voltage (+V) required to saturate the transistor switch



Soln :- know that ,
$$I_{C(sat)} = \frac{V_{CC}}{R_C} = \frac{10V}{1k} = 10mA$$

$$I_C = I_B \beta \Rightarrow I_B = \frac{I_{C(sat)}}{\beta} = \frac{10mA}{100} = 0.1mA$$

$$\therefore +V = I_B R_D + V_{BE}$$

$$= \{ (0.1 \times 47) + 0.7 \}$$

$$= 4.7 + 0.7 = 5.4V$$
 Ans

Galium

② A transistor is used as a switch. If $V_{CC} = 10V$, $R_C = 1K\Omega$, $I_{CBO} = 10\mu A$. determine the value of V_{CE} when transistor (i) cut off and (ii) saturated.

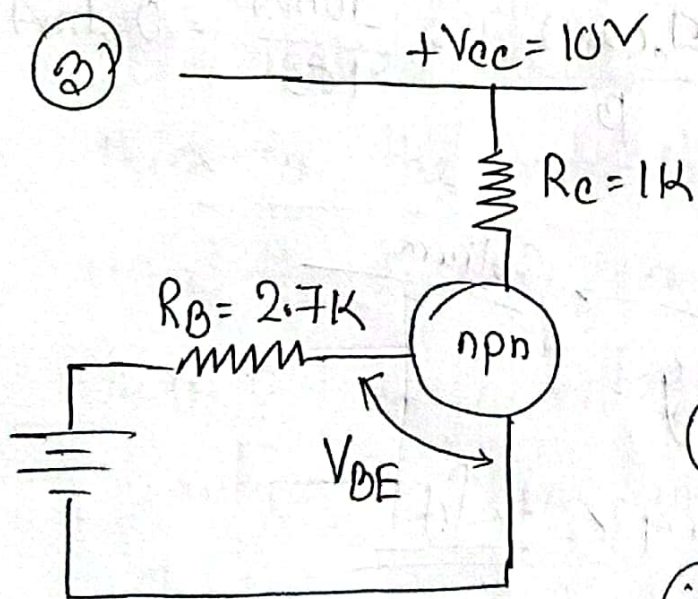
Soln: (i) At cut off

$$I_C = I_{CBO} = 10\mu A$$

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C \\ &= 10 - (10\mu A \times 1K\Omega) = 9.99V \quad \underline{\underline{Ans}} \end{aligned}$$

$$(ii) \quad I_C (sat) = \frac{V_{CC} - V_{Knee}}{R_C}$$

$$V_{CE} = V_{Knee} = 0.7V \quad \underline{\underline{Ans}}$$



$$V_{Knee} = 0.7V$$

$$V_{BE} = 0.7V$$

$$V_{BB} = 2V$$

(i) Minimum value of β for saturation.

(ii) If V_{BB} is changed to $1V$ and transistor has minimum $\beta = 50$ will transistor be saturated.

Soln: (i) $I_B = \frac{V_{BB} - V_{BE}}{R_B}$; $V_B = V_{BB} - V_{BE}$
 $= 0.48 \text{ mA}$

Now, $I_{e(sat)} = \frac{V_{ce} - V_{knom}}{R_c} = 9.2 \text{ mA}$

$\therefore \text{Minimum } \beta = \frac{I_{e(sat)}}{I_B} = \frac{9.2}{0.48} = 19.4$

(ii) $V_{BB} = 1 \text{ V}$, $\beta = 50$

$\therefore I_B = \frac{V_{BB} - V_{BE}}{R_B} = 0.111 \text{ mA}$

$I_c = \beta I_B = 50 \times 0.111 = 5.55 \text{ mA}$

Since, Collector current is less than saturation current ($I_c < I_{c(sat)}$), the transistor will not be saturated.

(4) In the astable multivibrator, $R_2 = R_3 = 10 \text{ k}\Omega$ and $C_1 = C_2 = 0.01 \mu\text{F}$. Determine time period and frequency of square wave.

Soln: $R = 10 \text{ k}\Omega = 10^4 \Omega$, $C = 0.01 \mu\text{F} = 10^{-8} \text{ F}$

$\therefore T = 1.4 RC = 0.14 \text{ msec}$ Ans

$\therefore f = \frac{1}{T_{\text{sec}}} = \frac{1}{1.4 \times 10^{-4}} = 7 \times 10^3 \text{ Hz}$ Ans

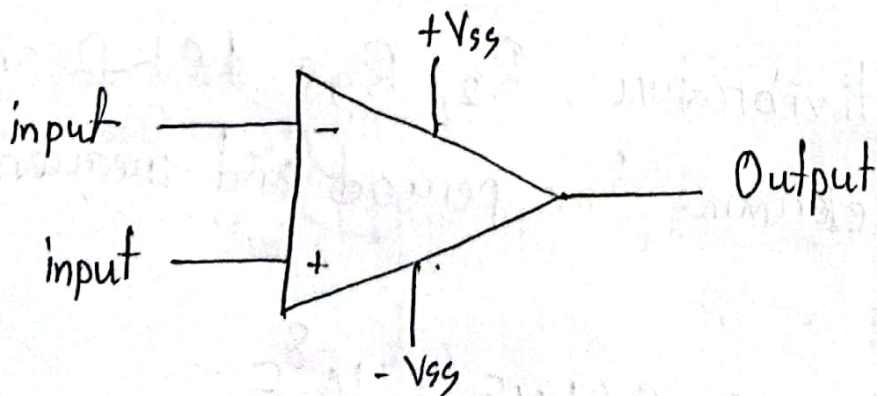
Segment 3

Operational Amplifier Fundamentals

- $\Rightarrow$ Introduction of operational Amplifiers
- $\Rightarrow$ Ideal Characteristics of an OP AMP
- $\Rightarrow$ Inverting OP AMP - up to gain calculation
- $\Rightarrow$ Non-inverting OP AMP - "
- $\Rightarrow$ Summing Amplifier
- $\Rightarrow$ Differentiator - integrator Amplifier - ~~DIFFERENTIAL~~

Operational Amplifier : — (OP-AMP)

An operational Amplifier is a circuit that can perform such mathematical operation as summing, subtraction, integration and differentiation.

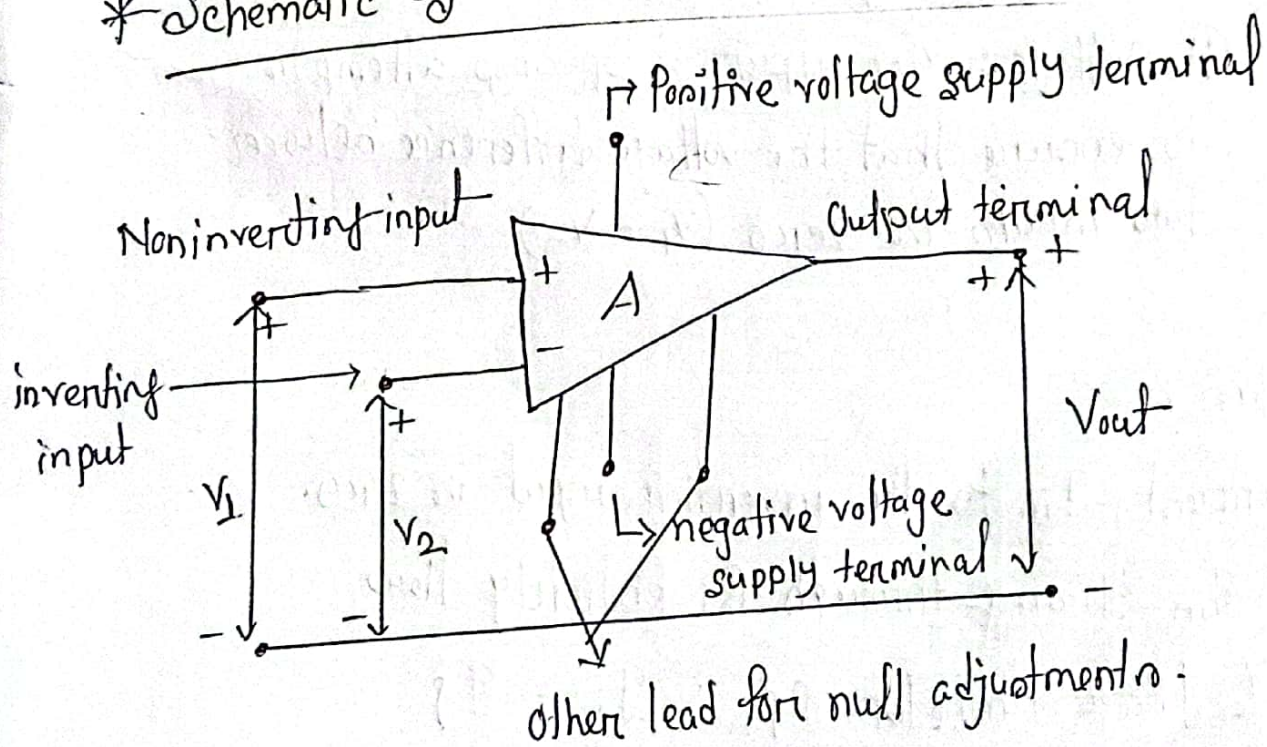


* Ideal Characteristics :-

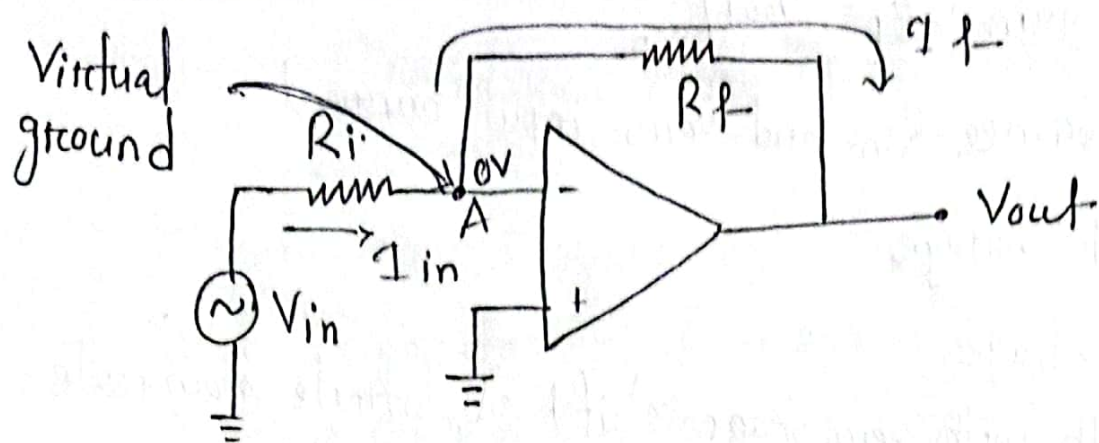
- (1) Infinite open-loop gain ; $G_T = V_{out}/V_{in}$
- (2) Infinite input impedance R_{in} and zero input current
- (3) Zero input offset voltage
- (4) Infinite output voltage range
- (5) Infinite bandwidth with zero phase shift & Infinite slew rate
- (6) Zero output impedance R_{out} .
- (7) Zero noise
- (8) Infinite power supply rejection ratio.

~~* Inverting OP-AMP - op to gain calculation :-~~

* Schematic symbol of OP-AMP :- Draw



* Inverting OP AMP - gain up to calculation :-



An OP-AMP can be operated as an inverting amplifier as shown in above circuit.

Inverting op-amps work —

- (1) The current :- No current flows into the inputs of the op-amp ($I_+ = I_- = 0$)
- (2) The Voltage :- The output of op-amp attempts to ensure that the voltage difference between two inputs is zero ($V_+ = V_-$)

Voltage gain :

The current I_{in} to the inverting input is zero. Therefore I_{in} flowing through R_i entirely flow through feedback resistor R_f $\therefore I_{in} = I_f$

Now, $I_{in} = \frac{\text{Voltage across } R_i}{R_i} = \frac{V_{in} - V_A}{R_i} = \frac{V_{in}}{R}$

and $I_f = \frac{\text{Voltage across } R_f}{R_f} = \frac{V_A - V_{out}}{R_f} = -\frac{V_{out}}{R_f}$

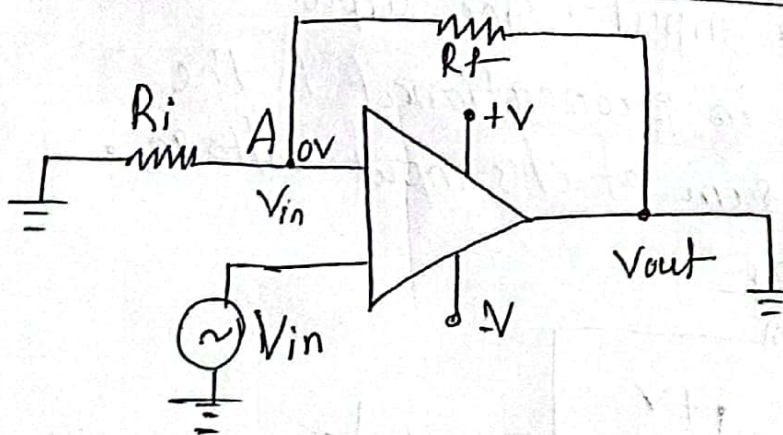
Since $I_{in} = I_f$

$$\therefore \frac{V_{in}}{R_i} = -\frac{V_{out}}{R_f}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

$\therefore$ Voltage gain, $A_{cl} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$

* Voltage gain in non-inverting op-amp :-



Voltage across $R_i = V_{in} - V_A = V_{in} - 0$

Voltage across $R_f = V_{out} - V_{in}$

Now current through R_i = current through R_f

$$\Rightarrow \frac{V_i - 0}{R_i} = \frac{V_{out} - V_{in}}{R_f}$$

$$\Rightarrow V_i R_f = V_{out} R_i - V_{in} R_i$$

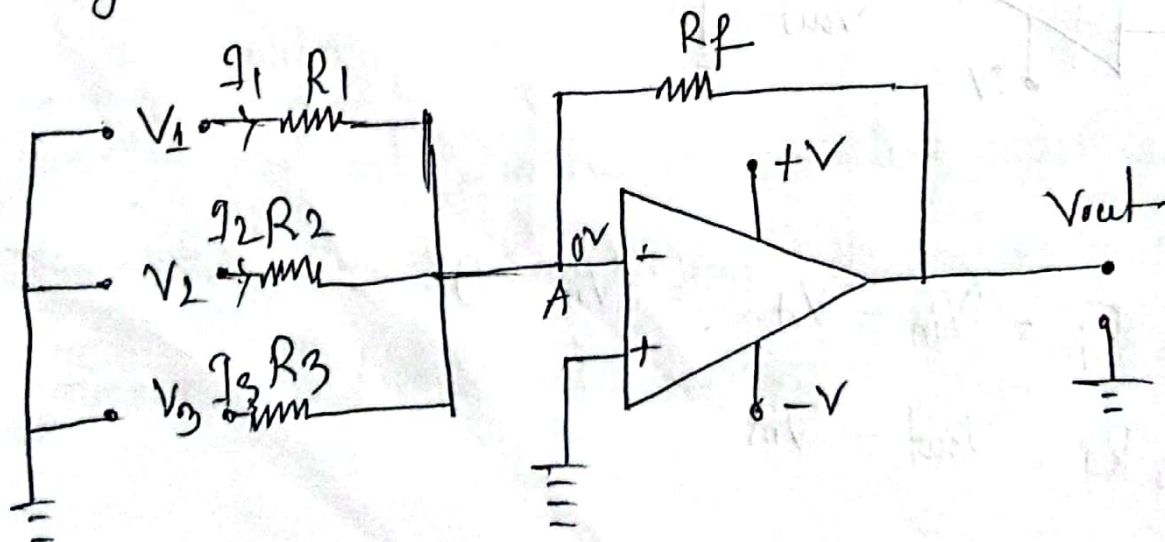
$$\Rightarrow V_i (R_f + R_i) = V_{out} R_i$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_i}$$

$$\therefore \text{Voltage gain, } A_{cl} = \frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_i} \quad \underline{\underline{A_1}}$$

* Summing Amplifiers:

A summing amplifier is an inverted OP-AMP that can accept two or more input. The output voltage of a summing amplifier is proportional to the negative of the algebraic sum of its input voltage.



$$\therefore I_f = I_1 + I_2 + I_3$$

$$\begin{aligned} \text{Output voltage, } V_{out} &= -I_f R_f = -R_f (I_1 + I_2 + I_3) \\ &= -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) \end{aligned}$$

$$\text{if } R_1 = R_2 = R_3 = R$$

$$\therefore V_{out} = -\frac{R_f}{R} (V_1 + V_2 + V_3)$$

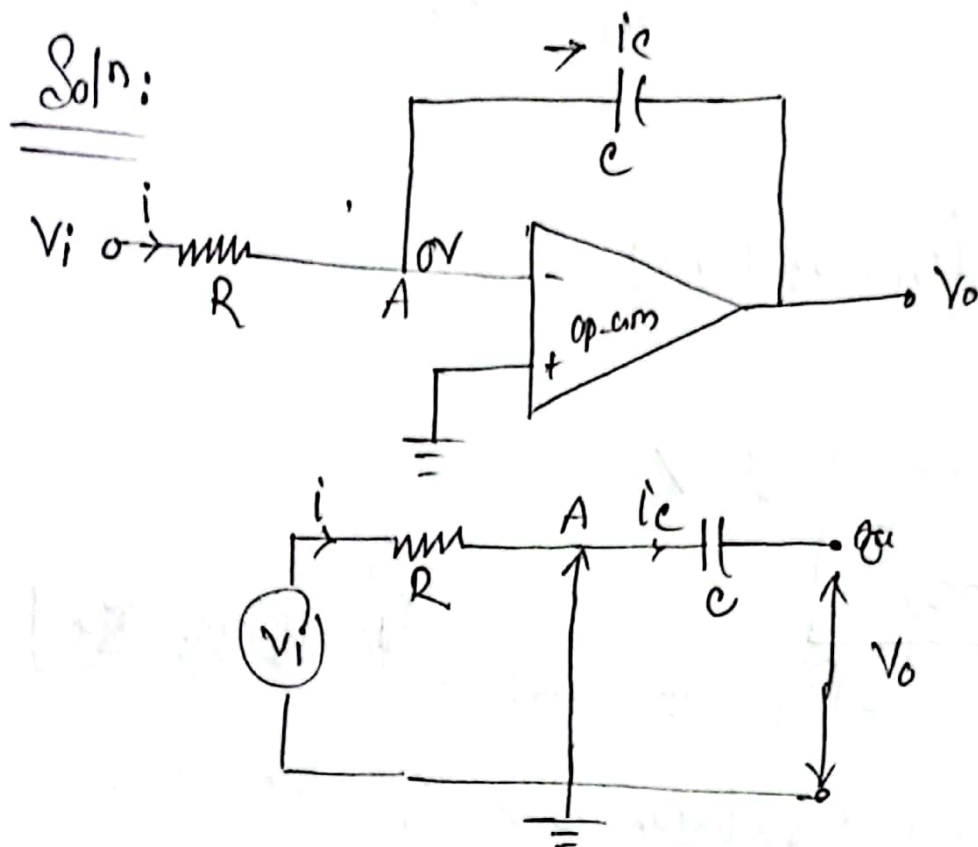
Gain of the amplifier is unity $\rightarrow$

$$\boxed{R_f = R (R_1 = R_2 = R_3)}$$

Output voltage, $V_{out} = -(V_1 + V_2 + V_3)$; ~~$[R_1 = R_2 = R_3 = R_f]$~~
 when the gain of summing amplifiers is unity, the output voltage is algebraic sum of the input voltage.

* Integral op-amp :-

⇒ the output is the integral of the input with an inversion and scale multiplier of $1/RC$.



Point A is at virtual ground, the virtual ground equivalent circuit of operational integrator. Because of virtual ground and infinite impedance of the OP-AMP, all of the input current i flows through the capacitor i.e. $i = i_c$

$$\text{Now, } i = \frac{V_i - V_A}{R} = \frac{V_i}{R} \quad \text{--- (i)}$$

Also, voltage across capacitor is $V_c = V_A - V_o = -V_o$

$$\therefore i_c = \frac{C dv_c}{dt} = -C \frac{dv_o}{dt} \quad \text{--- (ii)}$$

From eqn (i) & (ii) $\frac{v_i}{R} = -C \frac{dv_o}{dt}$

$$\Rightarrow \frac{dv_o}{dt} = -\frac{1}{RC} v_i \quad \text{--- (iii)}$$

To find the output, we integrate both sides of eqn (iii)

$$\int \frac{d}{dt} v_o = -\frac{1}{RC} \int v_i dt$$

$$\Rightarrow v_o = -\frac{1}{RC} \int v_i dt \quad \text{--- (iv)}$$

Eqn (iv) shows that the output is the integral of the input with an inversion and scale multiplier of $1/RC$

* Differentiator :-

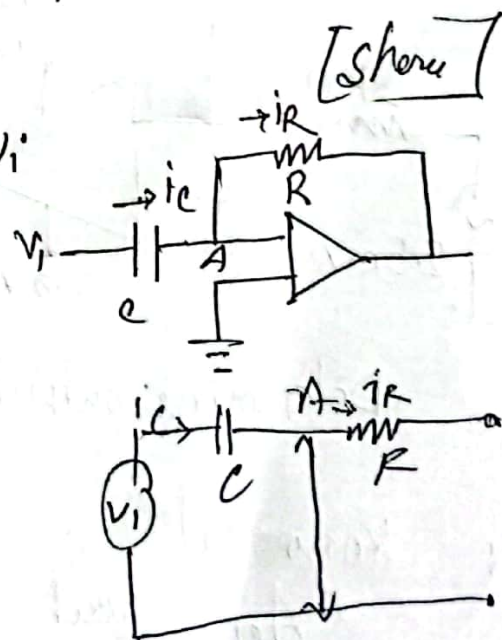
$$i_R = \frac{0 - v_o}{R} = -\frac{v_o}{R} \quad \text{and} \quad v_c = v_i - 0 = v_i$$

$$i_c = C \frac{dv_c}{dt} = C \frac{dv_i}{dt}$$

$$\therefore -\frac{v_o}{R} = C \frac{dv_i}{dt}$$

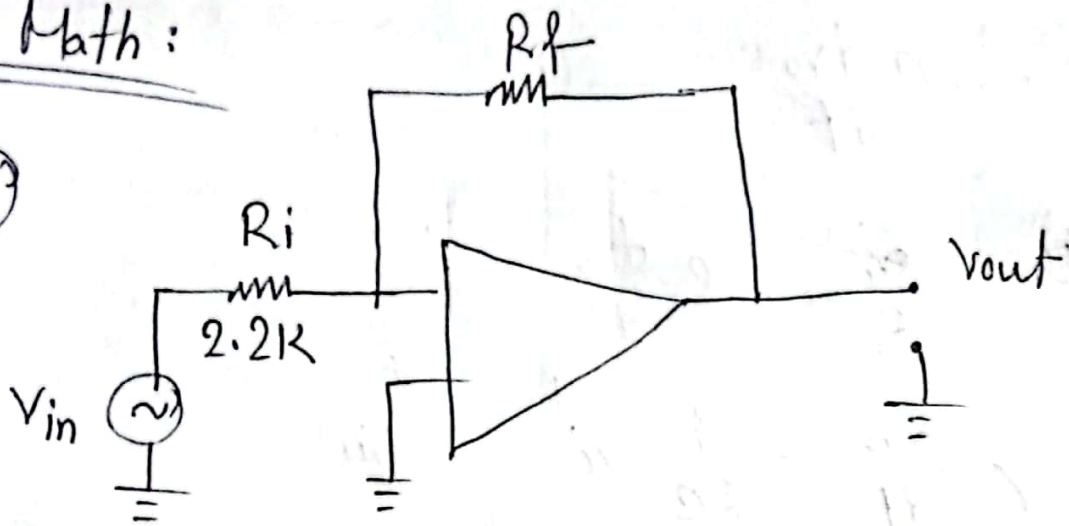
$$\Rightarrow v_o = -RC \frac{dv_i}{dt}$$

output is the differentiation of input with an inversion and scale multiplier of RC $i_c = i_R$



* Math:

(1)



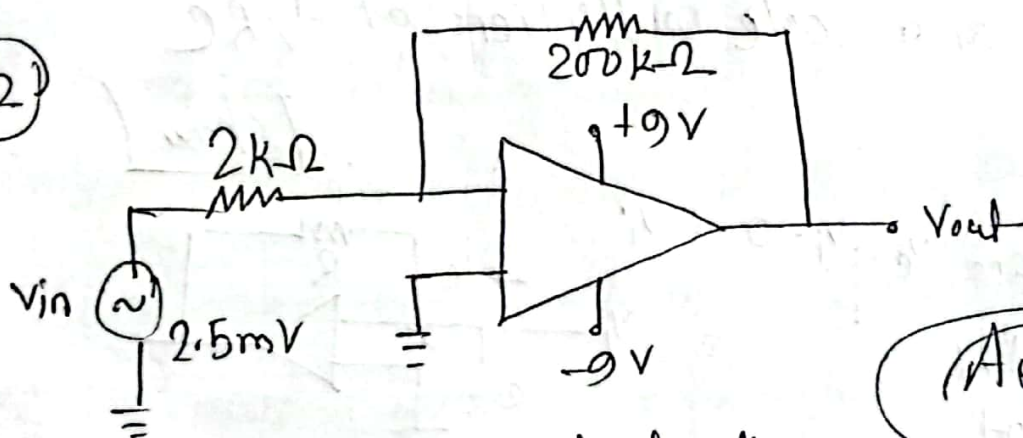
Determine the value of R_f required to produce a closed-loop voltage gain of -100 .

Soln: Know that,

$$A_{cl} = -\frac{R_f}{R_i}$$

$$\Rightarrow -100 \times -2.2k = R_f \Rightarrow R_f = 220k\Omega$$

(2)



Determine output voltage

$$A_{cl} = -100$$

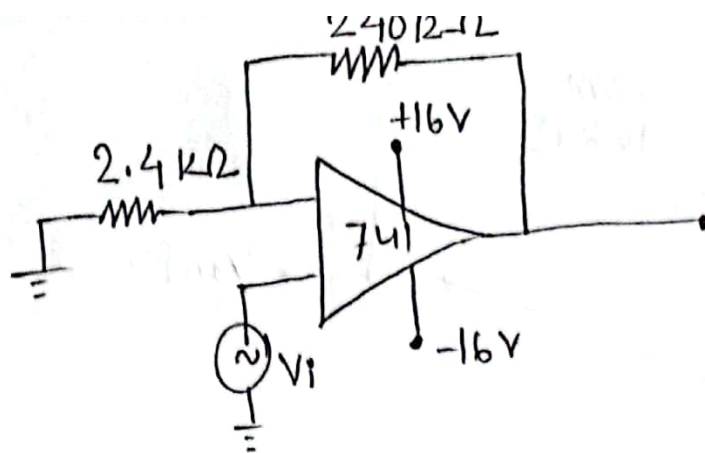
Soln: Know that

$$A_{cl} = \frac{V_{out}}{V_{in}}$$

$$; A_{cl} = -R_f/R_i = -100$$

$$\Rightarrow V_{out} = A_{cl} \times V_{in} = -250mV = -0.25V$$

(3)



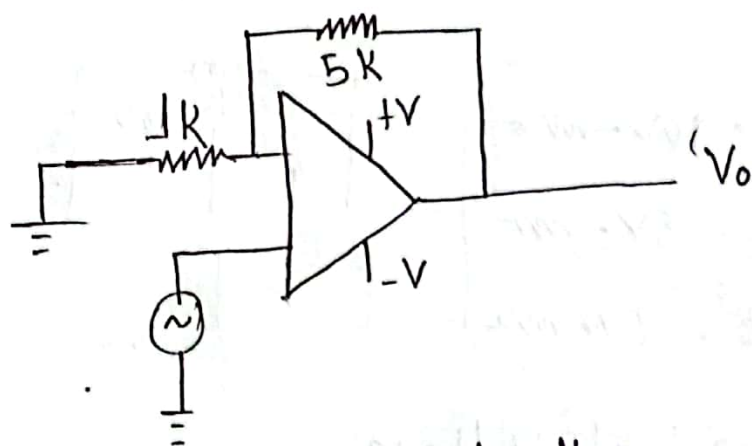
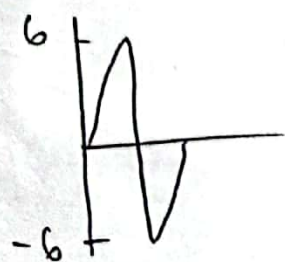
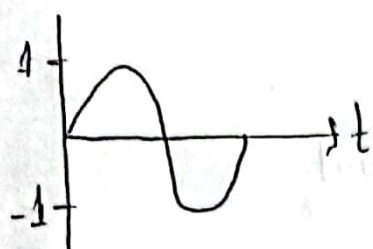
Output voltage
input of 120 μ V

Soln:- Voltage gain, $A_{cl} = V_{out}/V_{in}$

Again, $A_{cl} = 1 + R_f/R_i = 1 + 240/2.4 = 101$

$\therefore V_{out} = A_{cl} \times V_{in} = 101 \times (120 \mu V)$
 $= 12.12 \text{ mV}$ Ans

(4)



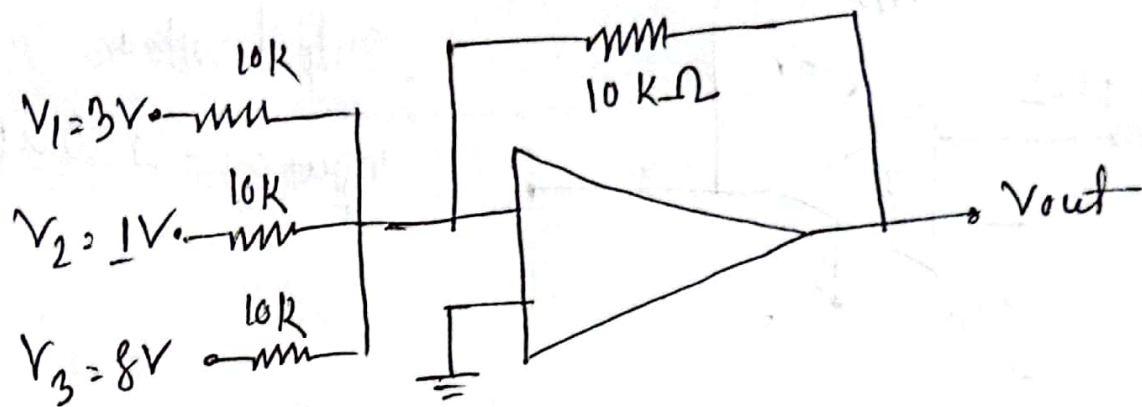
find peak-to-peak output voltage

Soln:- Input signal 2V peak to peak

Voltage gain $A_{cl} = 6$

$\therefore V_o = A_{cl} \times V_{in} = 6 \times 2 = 12$ Ans

5



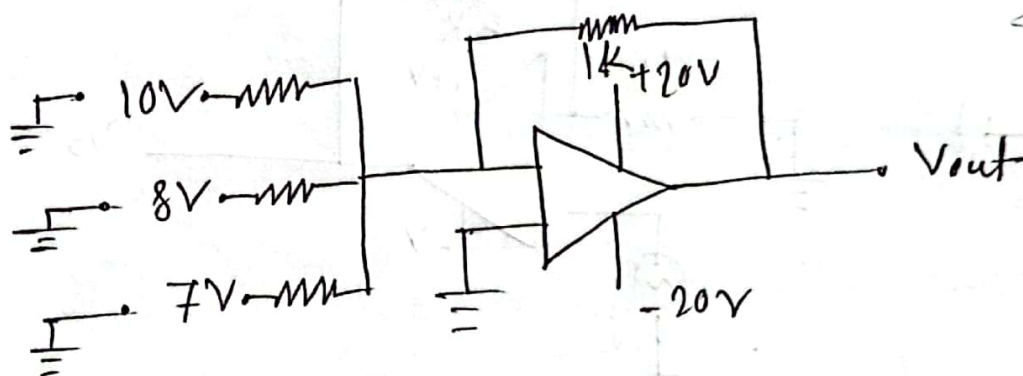
output voltage

Soln: gain of amplifier unity —

$$V_{out} = -(V_1 + V_2 + V_3)$$

$$= -12V \quad \underline{\underline{A}}$$

6



Output Voltage

Soln:

$$V_{out} = -\frac{R_F}{R} (V_1 + V_2 + V_3)$$

$$= -\frac{1}{10} (8 + 80 + 7)$$

$$= -2.5V \quad \underline{\underline{A}}$$

Segment - 4

OP. AMP Application

- $\Rightarrow$ Introduction of feedback. What is positive & negative feedback
- $\Rightarrow$ Principle of negative feedback in amplifier
- $\Rightarrow$ Derive the negative gain of negative feedback in amplifier
- $\Rightarrow$ Show that the input impedance of an amplifier increases due to negative voltage feedback
- $\Rightarrow$ Electronic oscillator & its essential component
- $\Rightarrow$ Hartley Oscillator [Details]
- $\Rightarrow$ Different type of transistor Oscillator ~~[Details]~~

Feedback :-

The process of returning part of the signal output from a circuit or device back to the input of that circuit or device.

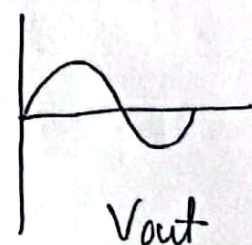
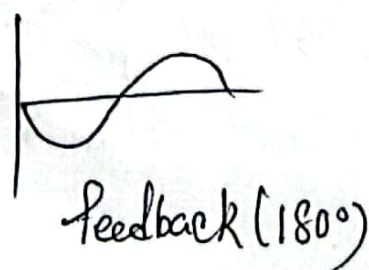
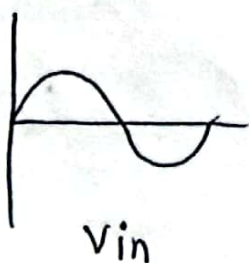
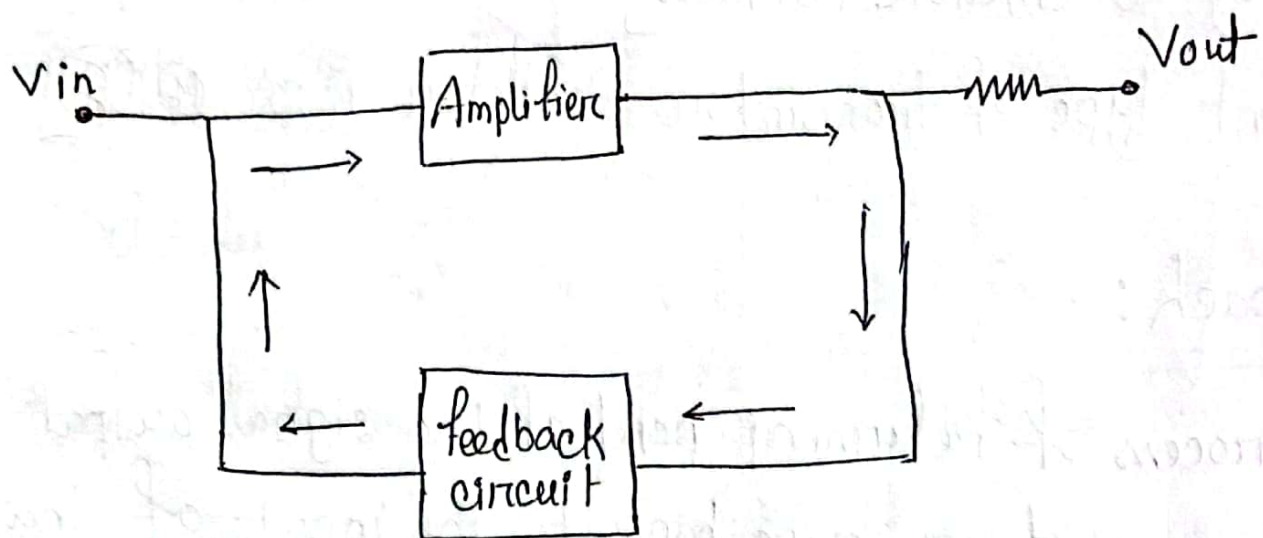
(i) positive feedback :-

When the feedback energy is in phase with the input signal and thus aids it, is called positive feedback.

(ii) Negative feedback :-

When the feedback energy is out phase with the input signal and thus opposes it, is called negative feedback.

* Principle of Negative feedback in amplifiers :-



A feedback amplifier has two parts, an amplifier and feedback. The feedback circuit usually consists of resistors and returns a fraction of output energy back to the input. Let's assume with illustrative, the output of the amplifier is $10V$. The fraction mv of this output i.e. 100 mV is feedback to input where it is applied in series with the input signal of 101 mV . As feedback is negative, therefore, only 1 mV appears at the input terminals of the amplifier.

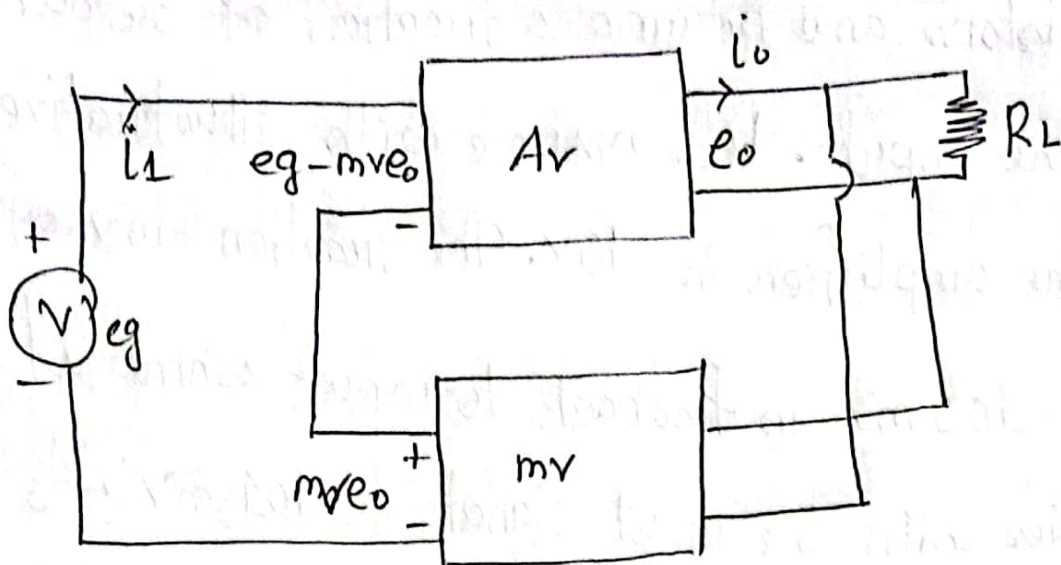
$$A_v = \frac{10}{1\text{ mV}} = 10,000 \text{ [without feedback]}$$

$$\text{fraction of output voltage feedback, } m_v = \frac{100\text{ mV}}{10V} = 0.01$$

$$\text{Gain of amplifier with negative feedback, } A_{vf} = \frac{10V}{101\text{ mV}} = 100$$

$\Rightarrow$ When negative voltage feedback applied, the gain of the amplifier reduced ($10,000 > 100$)

* Gain of negative feedback amplifier derive :-



input impedance :- let; Z_{in} without feed back and Z_{in}'

with feed back, input current is

$$\therefore e_g - m v_{e_o} = Z_{in} i_1$$

$$\Rightarrow e_g = (e_g - m v_{e_o}) + m v_{e_o}$$

$$= (e_g - m v_{e_o}) + m v A_v (e_g - m v_{e_o}); [e_o = A_v (e_g - m v_{e_o})]$$

Consider,

The gain of amplifier without feedback is A_v .

Negative feedback is then applied by feeding a fraction $m v$ of the output voltage e_o back to amplifier input.

Therefore the actual input to the amplifier is the signal voltage e_g minus feedback voltage $m v_{e_o}$

Actual input to amplifier = $e_g - m_v e_o$

The output e_o must be equal to the input voltage ($e_g - m_v e_o$) multiplied by gain A_v of the amplifier

$$e_o = A_v (e_g - m_v e_o) \\ = A_v e_g - A_v m_v e_o$$

$$\Rightarrow A_v e_g = e_o (1 + A_v m_v)$$

$$\Rightarrow A_v \frac{e_g}{e_o} = (1 + A_v m_v)$$

$$\Rightarrow \frac{e_o}{e_g} = \frac{A_v}{1 + A_v m_v}$$

e_o/e_g is the voltage gain of amplifier with feedback.

$$\therefore A_{vf} = \frac{A_v}{1 + A_v m_v} \quad [\text{Showered}]$$

So, it is clear that by applying negative feedback, input impedance of amplifier is increased by a factor $1 + A_v m_v$. So, the input impedance has increased. [showed]

As for output due to negative feedback, output impedance of amplifier is decreased by a factor $(1 + A_v m_v)$ [showed]

Oscillator :- [Positive feedback Amplifier]

Oscillator is a mechanical or electronic device that work on the principles of oscillation and produces a continuous, repeated, alternating waveform.

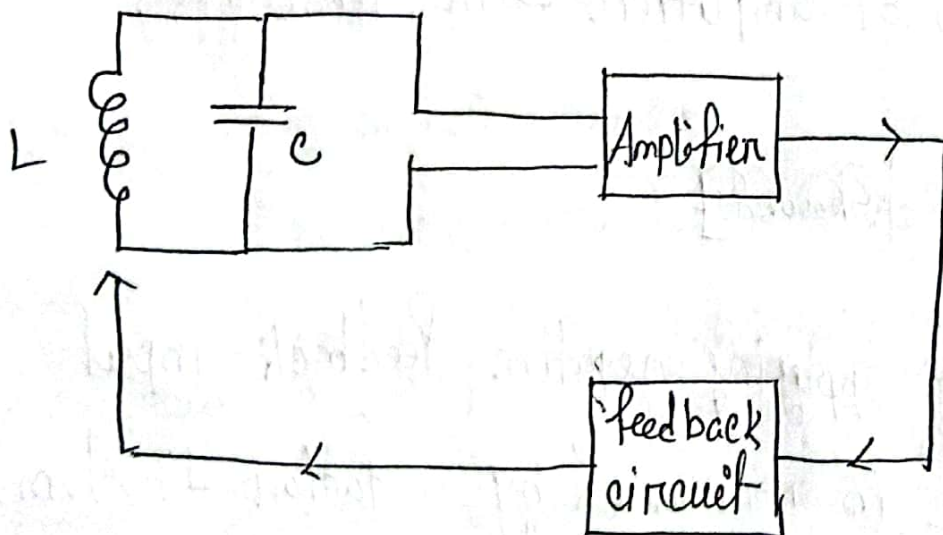
Oscillator essential components are

- ① tank circuit
- ② Amplifier
- ③ feedback circuit

frequency of oscillations

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

operating frequency

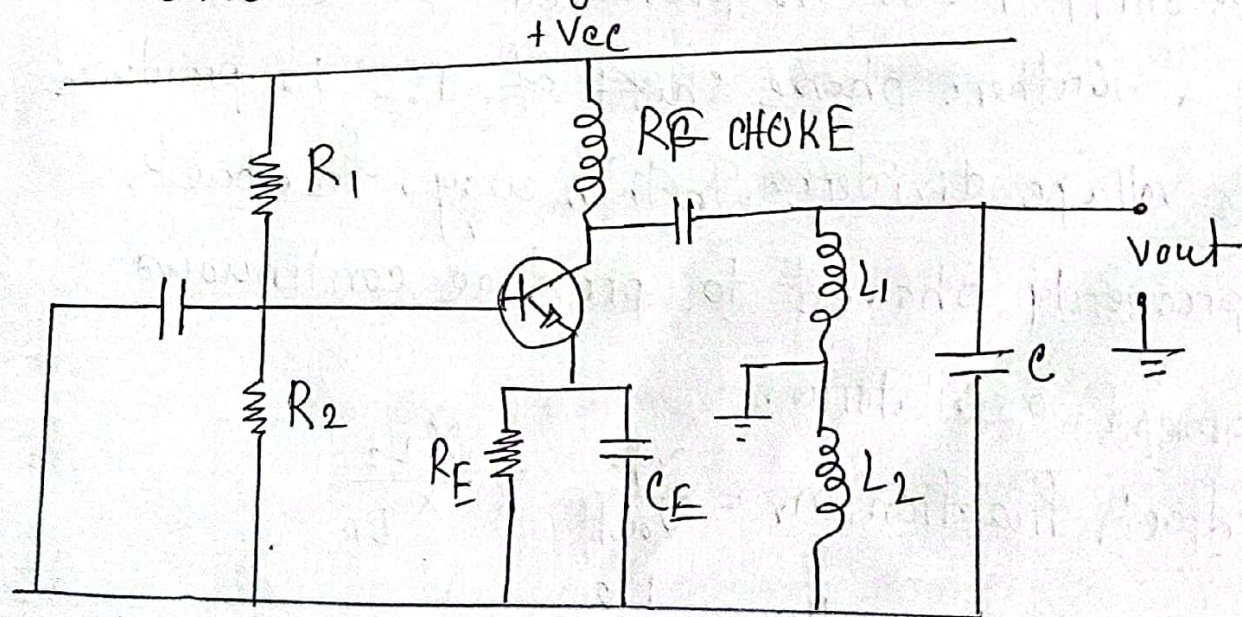


* Different type of transistor Oscillators:-

- ① Tuned collector oscillator
- ② Colpitt's oscillator
- ③ Hartley oscillator
- ④ Phase shift oscillator
- ⑤ Wien Bridge oscillator
- ⑥ Crystal oscillator

* Hartley Oscillator :- [Ralph Hartley → 1915]

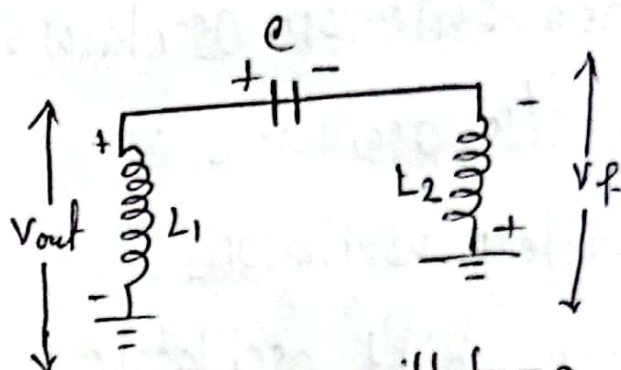
The Hartley oscillator is an electronic oscillator circuit in which oscillation frequency is determined by a tuned circuit consisting of capacitors & inductors.



Operation/Working process

When the circuit is turned on, the capacitor is charged, when the capacitor is fully charged, it

discharges through L_1 and L_2 setting up oscillations of frequency determined by *exp.. The output voltage of the amplifier appears across L_1 and feedback



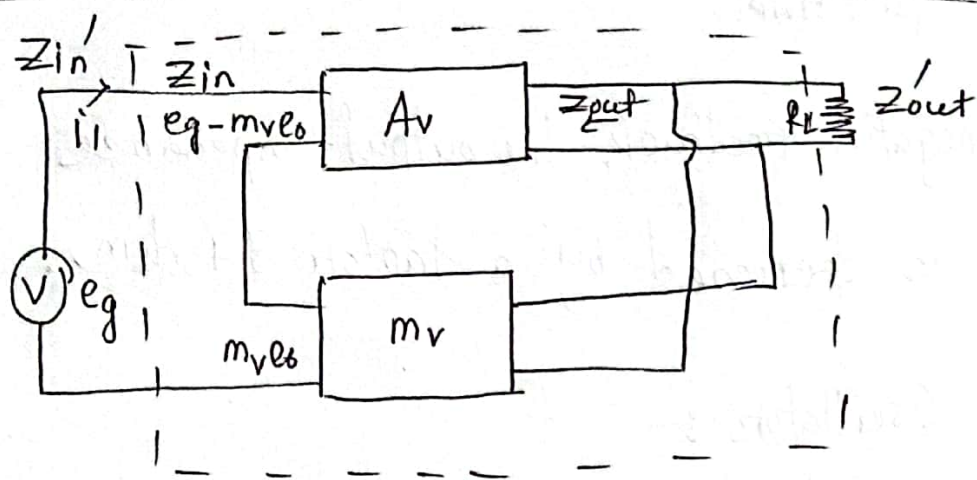
voltage across L_2 . The voltage across L_2 is 180° out of phase with the voltage developed across L_1 (v_{out}). It is easy to see that voltage feedback to the transistor provides positive feedback. A phase shift of 180° is produced by the transistor and a further phase shift of 180° is produced by the L_1 - L_2 voltage divider. In this way, feedback is properly phased to produce continuous undamped oscillations.

$$\text{feedback fraction } m_v = \frac{v_f}{v_{out}} = \frac{X_{L_2}}{X_{L_1}}$$

$$m_v = \frac{L_2}{L_1}$$

Extra:

* Input/Output impedance due to negative feedback: -



input let, Z_{in} without feedback, Z'_{in} with feedback.
input current i_1 .

Know that, $e_g - m_v e_o = i_1 Z_{in}$

$$\Rightarrow e_g = i_1 Z_{in} + m_v e_o$$

$$= (e_g - m_v e_o) + m_v e_o$$

$$= (e_g - m_v e_o) + m_v A_v (e_g - m_v e_o); e_o = A_v (e_g - m_v e_o)$$

$$= i_1 Z_{in} (1 + A_v m_v)$$

$$\Rightarrow \frac{e_g}{i_1} = Z_{in} (1 + A_v m_v)$$

$$\Rightarrow Z'_{in} = Z_{in} (1 + A_v m_v); e_g / i_1 = Z'_{in}$$

So it is clear that by applying negative feedback, input impedance of the amplifier is increased by a factor $1 + A_v m_v$.

output

$$Z_{out}' = \frac{Z_{out}}{(1 + A_{v_{mv}})}$$

By applying negative feedback, the output impedance of amplifier is decreased by a factor $1 + A_{v_{mv}}$.

* Sinusoidal Oscillator :-

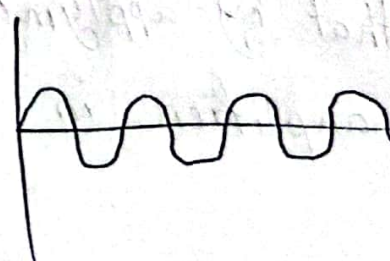
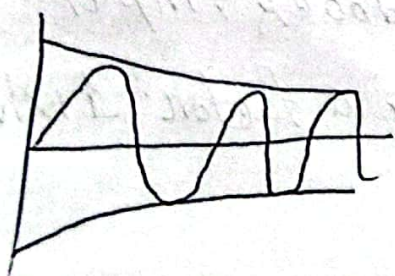
An electronic device that generates sinusoidal oscillation of desired frequency is known as sinusoidal oscillator.

Type $\rightarrow$

① Damped Oscillation ② Undamped oscillations

A circuit which produces electrical oscillation of any desired frequency is known as an oscillatory circuit / Tank circuit

it creat with inductors and capacitor.



* Math

- ① The overall gain of a multistage amplifier is 140. When negative voltage feed-back is applied, the gain is reduced to 17.5. find the fraction of the output that is feedback to the input.

Solⁿ: $A_v = 140$, $A_{vf} = 17.5$

$$\therefore A_{vf} = \frac{A_v}{1 + A_v m_v}$$

$$\Rightarrow m_v = \frac{140 - 17.5}{2450} = \frac{1}{20} \cancel{A_1}$$

- ② When negative feedback is applied to an amplifier of gain 100, the overall gain falls to 50.

i) Calculate the fraction of the output voltage feedback.

ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

Soln: $A_v = 100$ $A_{vf} = 50$

(i) $A_{vf} = \frac{A_v}{1 + A_v m_v}$

$\Rightarrow m_v = \frac{100 - 50}{5000} = 0.01 \underline{A}$

(ii) $A_{vf}' = 75$; $m_v = 0.01$; $A_v' = ?$

$A_{vf}' = \frac{A_v'}{1 + A_v' m_v}$

$\Rightarrow A_v' = \frac{75}{1 - 0.75} = 300 \underline{A}$

(3) The gain of an amplifier without feedback is 50 whereas with negative voltage feedback it falls to 25. If due to ageing, the amplifier gain falls to 40. Find the percentage reduction in stage gain (i) without feedback (ii) with negative feedback.

Soln: $A_{vf} = \frac{A_v}{1 + A_v m_v}$

$\therefore m_v = 1/50$

(i) Without feedback:- The gain of the amplifier without feedback is 50. However, due to ageing it falls to 40

$$\% \text{ age reduction in stage gain} = \frac{\text{Amplifier ~~without~~ feedback} - \text{with ageing feedback fall}}{\text{Amp ~~without~~ feedback}} \times 100$$

$$\therefore \% \text{ age reduction in stage gain} = \frac{50 - 40}{50} \times 100\%$$

$$= 20\% \quad \underline{\underline{A}}$$

(ii) With negative feedback: When the gain without feedback was 50, the gain with negative feedback was 25

$$\therefore A_{vf}' = \frac{A_v}{1 + A_v m_v} = 22.2$$

$$\therefore \% \text{ age reduction in stage gain} = \frac{25 - 22.2}{25} \times 100\%$$

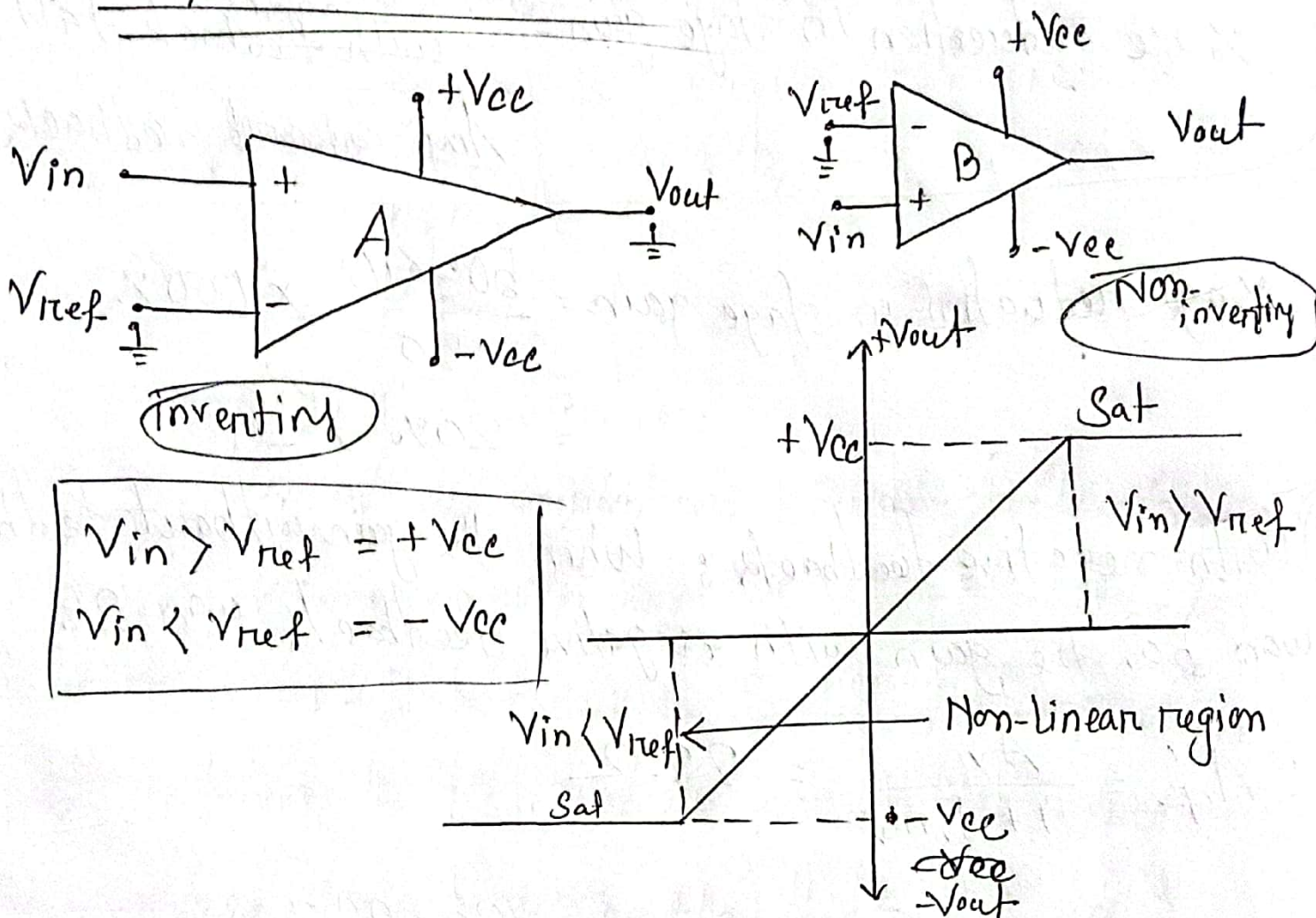
$$= 11.2\% \quad \underline{\underline{A}}$$

Segment - 5

Non-linear Application

⇒ Comparators, Precision Rectifiers

Comparator Circuit :-



It is an electronic decision making circuit that makes use of an operational amplifier's very high gain in its open-loop state [There is no feedback resistor].

inverting

It produces one of the two values, $+V_{sat}$ and $-V_{sat}$ at output based on the value of input V_{in} and reference V_{ref} .

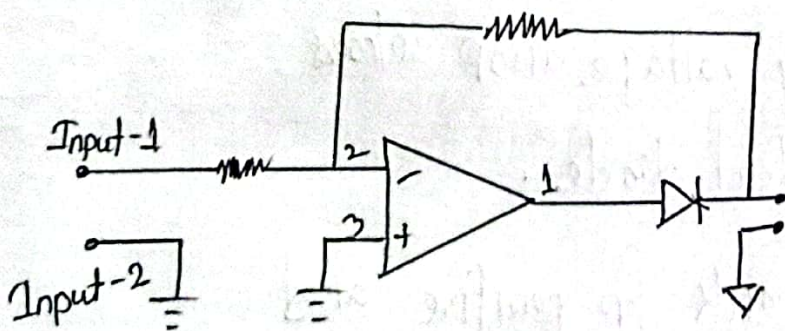
- The output $-V_{sat}$ if V_{in} is greater than V_{ref}
- The output $+V_{sat}$ if V_{in} is less than V_{ref}

Non-inverting

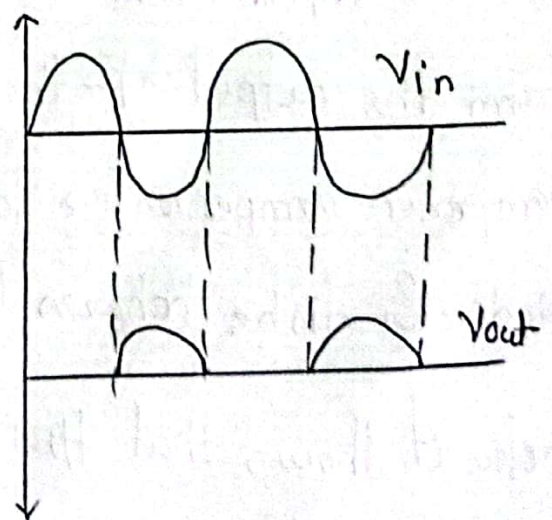
It produces one of the two value, $+V_{sat}$ and $-V_{sat}$ at output based on the value of input V_i and reference V_{ref} .

- The output $+V_{sat}$ if V_{in} greater than V_{ref}
- The output $-V_{sat}$ if V_{in} is less than V_{ref} .

▣ Precision Rectifier :-



(Fig:1)



~~It converts AC to DC~~

A rectifier is a circuit that converts alternating current to direct current. By using diodes rectifier method, it can only used if the input voltage to the circuit is greater than forward voltage of diode ($0.7V$)

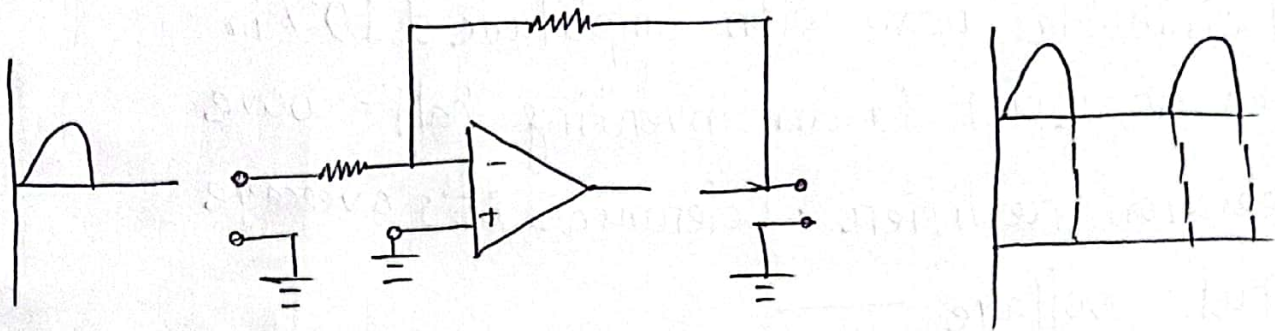
To overcome this issue, we can use op-amp rectifier which is also known as precision rectifier. With this, we are not losing the $0.6V/0.7V$ voltage drop across the diode.

Half-wave precision rectifier : —

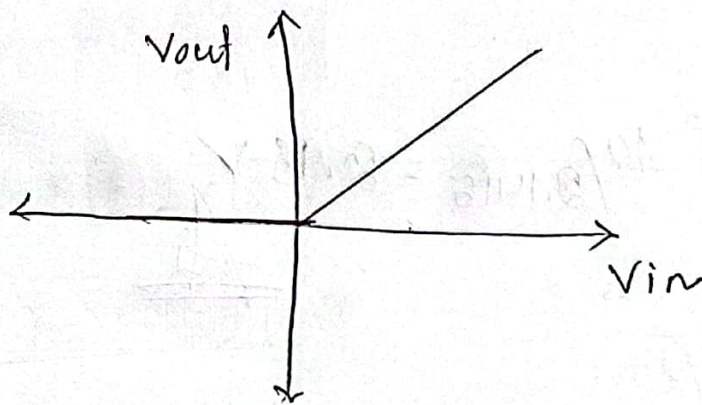
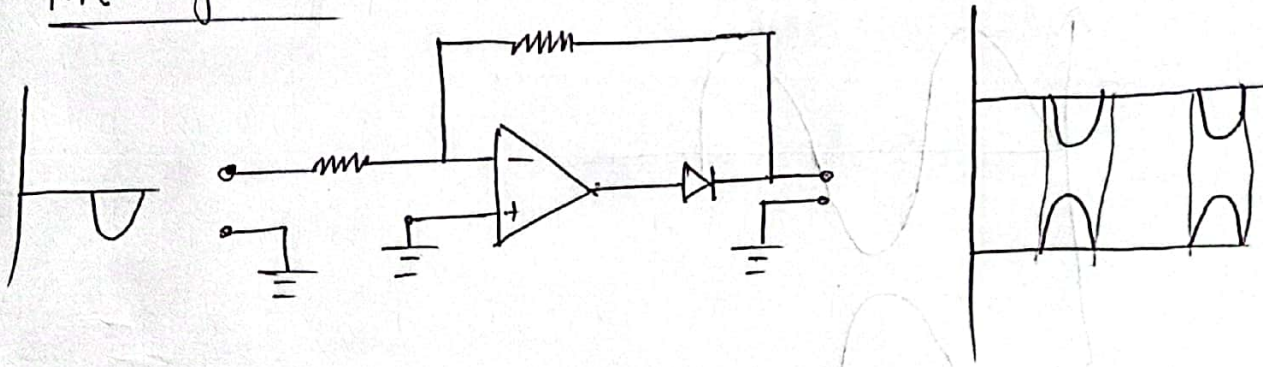
from fig-1 it shows that input and output waveform of the precision rectifier, which is exactly equal to the input. That because we are taking the feedback from the output of the diode and op-amp compensates for any voltage drop across diode. So diode behaves like ideal diode.

below it shows that How it work for positive and negative half cycle —

for positive:



for negative:

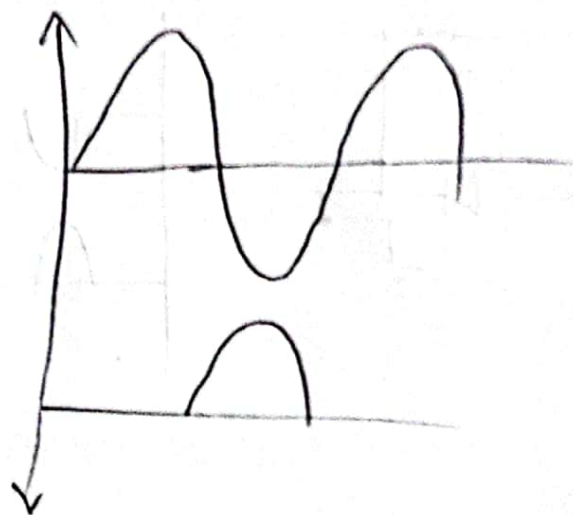


Transfer Characteristics.

* Math

- ① A sinusoidal wave with amplitude $\pm 10V$ is given as input to an inverting half-wave precision rectifier. Determine its average output voltage —

Solⁿ:

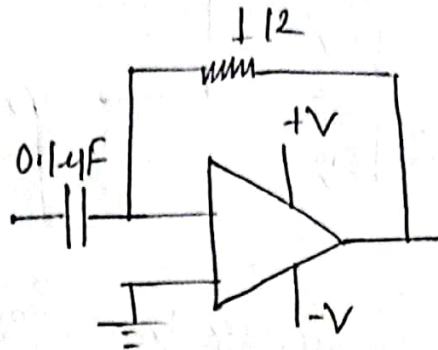
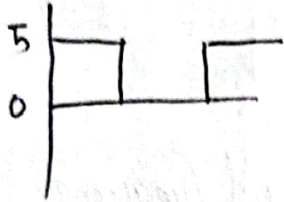


$$V_{av} = V_m / \pi = 10 / 3.1416 = 3.18 V$$

Ans

Some Extra Math

①



The square wave input to a differentiator circuit. find output voltage if input goes from 0V to 5V in 0.1ms

Solⁿ: We know that,

$$V_o = -RC \frac{dV_i}{dt}$$

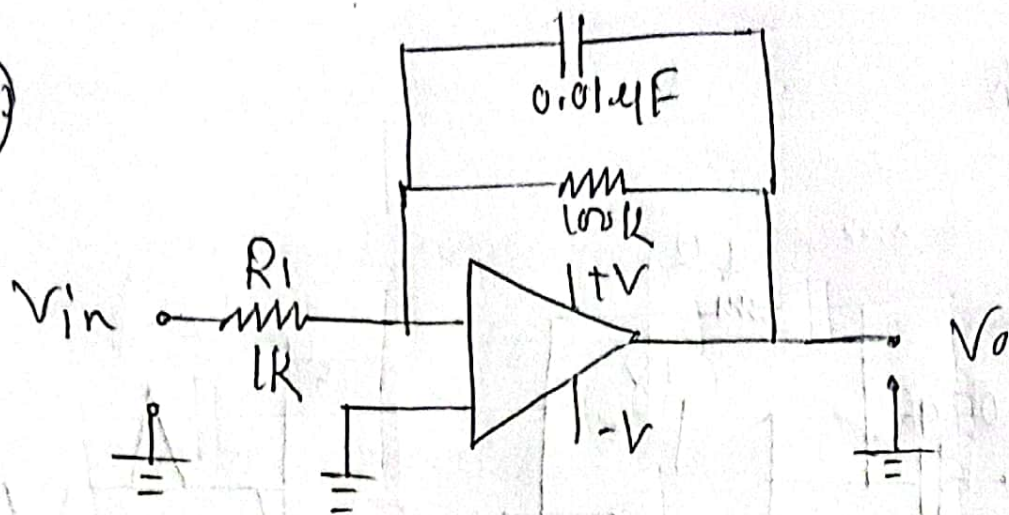
$$\therefore R = 1k \times (0.1 \times 10^{-6}) = 0.1 \times 10^{-3} s$$

Also Also, $\frac{dV_i}{dt} = \frac{5V - 0V}{0.1ms} = \frac{5V}{0.1ms} = 5 \times 10^4 V/s$

$$\therefore V_o = - (0.1 \times 10^{-3}) (5 \times 10^4) = -5V$$

AI

(2)



Determine the lower frequency limit (critical frequency) for the integrator circuit

Soln:

$$f_c = \frac{1}{2\pi R_f C}$$

$$R_f = 100k\Omega = 10^5\Omega, C = (0.01 \times 10^{-6})F$$

$$\therefore f_c = \frac{1}{2\pi \times (10^5) \times (0.01 \times 10^{-6})}$$

$$= 159 \text{ Hz}$$

③ A 1pF capacitor is available. Choose the inductor value in Hartly Oscillator so that $f = 1\text{MHz}$ and $m_v = 0.2$.

Soln:

feedback fraction $m_v = \frac{L_2}{L_1}$

$$\Rightarrow 0.2 = L_2/L_1$$

$$\Rightarrow L_1 = 5L_2$$

Again

$$f = \frac{1}{2\pi\sqrt{L_1 C}}$$

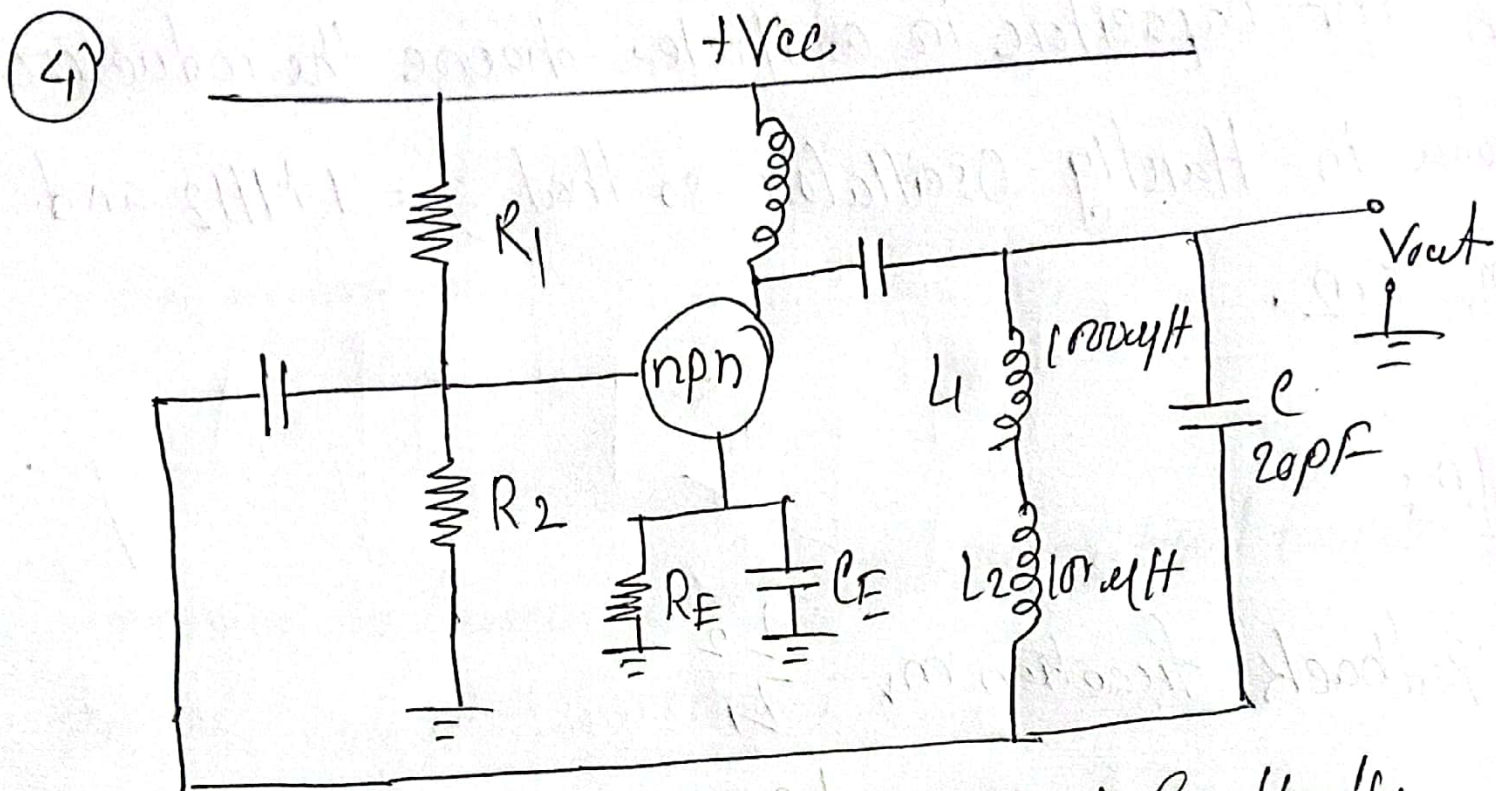
$$L_1 = \frac{1}{C(2\pi f)^2} = 25.3 \times 10^{-3}\text{H} = 25.3\text{mH}$$

$$\therefore L_1 + L_2 = 25.3\text{mH}$$

$$\Rightarrow 5L_2 + L_2 = 25.3$$

$$\Rightarrow L_2 = 25.3/6 = 4.22\text{mH}$$

$$\text{And } L_1 = 5L_2 = 21.1\text{mH}$$



i) Operating frequency ii) feedback for Hartly oscillator - The mutual inductance between coils
 $M = 20 \mu H$

Soln: i) $L_1 = 1000 \mu H$; $L_2 = 100 \mu H$
 $M = 20 \mu H$

$$\text{Total inductance } L_T = L_1 + L_2 + 2M$$

$$= 1140 \mu H = 1140 \times 10^{-6} H$$

$$\text{Capacitance } C = 20 pF = 20 \times 10^{-12} F$$

$$\begin{aligned}\therefore \text{Operating frequency } f_r &= \frac{1}{2\pi\sqrt{L_T C}} \\ &= 1052 \times 10^3 \text{ Hz} \\ &= 1052 \text{ kHz} \quad \underline{\underline{A1}}\end{aligned}$$

$$\begin{aligned}\textcircled{\text{ii}} \text{ feedback fraction } m_v &= \frac{L_2}{L_1} \\ &= 0.1 \quad \underline{\underline{A1}}\end{aligned}$$